State Capacity, Brokers, and Service Delivery

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Abstract

Social protection programs are central to people's lives. However, administrative burdens in accessing them create informal markets for brokers who help applicants navigate the process for a fee. Whether these brokers substitute for public capacity or complement it remains unclear. We study this question using a reform in an Indian state that expanded staffing at public service centers to help citizens submit applications, without increasing the number of officials deciding whether to approve them. Using pre-existing differences in district exposure to the reform, we find that a 1 percent increase in staffing raises broker-mediated applications by 1.7 percent, compared to 1.1 percent for direct applications. Yet direct applications become faster, reducing brokers' average speed advantage. These patterns rule out a model where public workers purely substitute for brokers and, under maintained assumptions, one where brokers only obtain priority through speed money. A model where public workers make brokers more productive can explain joint results. Our findings highlight the role of informal markets in building state capacity.

Keywords: State capacity; bureaucratic intermediaries; public service delivery; administrative burdens; informal markets; India.

JEL codes: D73, H11, O17.

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1. Introduction

Social protection programs are central to people's lives. A key factor contributing to the gap between public programs and their implementation is weak state capacity. This is the ability of a state to raise revenue, monitor implementation of programs, penalize corruption, and ensure universal access to basic public services (Dasgupta and Kapur, 2020; Muralidharan et al., 2023; Muralidharan, 2024). In India, the share of population covered by at least one social protection program grew from 19% in 2015 to 64% in 2025[1], while the number of public officials per 1000 people remaining one of the lowest in the world. Administrative burdens in accessing these programs often create informal markets for brokers who help applicants navigate the application process for a fee. They are unregulated private intermediaries who charge for help to complete paperwork, put together an application package, or make repeat visits to offices. There is a large political science literature that has documented the role of brokers in bureaucracies and politics in India, Latin America, and Sub-Saharan Afria (Auerbach and Thachil, 2023; Hassan et al., 2024; Auerbach and Kruks-Wisner, 2020).

Despite their ubiquitous presence in routine public life, especially when accessing basic public services, there is no consensus about their productivity or welfare implications. For instance, are they substitutes or complementary to the state? Brokers can save time, reduce application mistakes, and help citizens access public programs (Bettinger et al., 2012; Finkelstein and Notowidigdo, 2019; Gupta, 2026). However, they may also help clients bypass procedures or obtain preferential treatment from officials even leading to misallocation of public resources (Bertrand et al., 2007). Assistance can reduce the real costs of accessing the state, while preferential treatment can benefit applicants who can afford to pay for it at the expense of others, introducing efficiency-equity tradeoffs. Understanding access to the state therefore requires understanding both how citizens, brokers, and public officials interact in the delivery of public services.

A common policy view is: brokers exist and are widely used because governments make public services difficult to access directly[2]. It makes sense to think that citizens hire brokers when they cannot easily get help from public officials because they are already overloaded with work (Dasgupta and Kapur, 2020). This is especially true for countries with low levels of state capacity like India (Muralidharan, 2024). Policymakers therefore justifiably worry that access to basic public programs might become regressive in practice (Romero and Singh, 2022) and governments invest in personnel or infrastructure to make access to public programs easier. A natural expectation is that expanding state capacity should crowd out at least part of the private broker market.

[1] Our World In Data: Share covered by at least one social protection benefit

[2] This view is explicit in many administrative reforms. For instance, India's Direct Benefit Transfer system was designed to send welfare benefits directly to recipients and reduce intermediaries and leakage. Similarly, government guidance for Jan Seva Kendras describes application procedures as being designed so citizens can complete them without the help of "touts or middlemen."

We test if this is true by using a statewide staffing reform in a large Indian state. Citizens in our setting apply for government-issued certificates that establish income, caste, residence, birth, death, and other certificates. These documents are required to access welfare programs and public benefits. Citizens submit their applications through a statewide network of public service centers known as *Sewa Kendras* (SKs). In March 2023, the government sharply increased staffing at these centers. The reform expanded the state's capacity to help citizens prepare and submit applications, but it did not increase the capacity of the officials who reviewed and approved them (approvers). The process outline is shown in Figure 1.

Ex-ante the effects of such a reform on efficiency of public service provision is ambiguous. Higher frontline operators might have no effect on the time taken to process applications if there was slack in the approver stage. They may even worsen processing time if the inflow of applications become large enough to congest the approval stage. Or, even we consider a rise in frontline operators to improve effective approval capacity, say because approvers now receive better-quality applications, one may think that this might even improve average processing time time of applications!

We collaborate with the department of governance reforms of a large Indian state (32M people) and access their administrative records to observe how the state delivers public services and the private broker market. For every application, we observe submission and issuance dates, service type, location, and an associated mobile number. Most mobile numbers appear only once, which is intuitive because citizens typically need these certificates once in their lives, or once every 5-10 years. We observe a small set of mobile numbers recur across applications submitted for different citizens and use this repeated activity to identify likely intermediaries. In our baseline measure, we classify a mobile number as a broker if it appears on at least three applications during our entire November 2022-August 2023 sample. Because citizens typically need these certificates only occasionally, frequent reuse of the same number is unusual and is also viewed by government IT officials as a marker of intermediary activity. This measure will miss brokers who submit applications using their clients' own numbers and may misclassify shared family numbers or unpaid helpers as brokers. We interpret it as a proxy instead of a complete census of brokerage. If these classification errors are unrelated to application outcomes, they will generally attenuate measured differences between brokered and direct applications.

The statewide hiring expansion allocated new staff to existing office infrastructure, generating larger increases in application-stage capacity in some districts over others. We exploit this variation by interacting pre-reform staffing per capita with the subsequent statewide increase in hiring. Districts with greater pre-reform staffing were more exposed to the common expansion because the government allocated new operators through the existing service-center infrastructure. We use this exposure measure to instrument for actual staffing per capita and compare changes in applications, broker activity, and processing outcomes across more- and less-exposed districts. The

design controls for district fixed effects and common seasonal patterns. Identification requires that, absent the reform, districts with different exposure would not have experienced systematically different changes in these outcomes. We test this assumption by asking whether exposure predicts differential trends before the staffing expansion and by examining other policy changes that could have affected application demand or brokerage during the same period.

Contrary to the simple crowd-out prediction discussed above, we find that additional application-stage staffing expands the broker market disproportionately. A 1 percent increase in staffing raises broker-mediated applications by approximately 1.7 percent, compared to 1.09 percent for applications filed directly by citizens. This is not simply greater activity among existing brokers. The number of potential active brokers also rises with staffing, consistent with entry into the market. At the same time, direct applications increase. Thus, the reform expands access through both channels rather than merely shifting applications from one to the other. The larger response through brokers raises their share of application activity. Greater public capacity therefore crowds in both the use and the supply of private intermediation.

This result would have been intuitive if brokered applications retained their processing speed advantage, which they do not, motivating us to rethink the role of the private broker market. We find no change to processing time of brokered applications while citizen-submitted direct applications speed up and nearly close a significant pre-existing gap. It is therefore puzzling that broker use continues to rise with the observed processing-time advantage shrinking. Not only is it puzzling, it is also suggestive of other reasons for why brokers exist, beyond public workers being overworked and thus hard to reach by regular citizens. It motivates a theory of complementarities where additional public workers and brokers might make each other more productive. That is, frontline operators can reduce the time it takes for brokers to submit applications, while brokers can save frontline operators' time in helping prepare citizen applications. In such a world, it makes intuitive sense that brokers can expand even as direct service improves and the recorded processing-time advantage of using a broker falls.

To formalize the intuition of complementarities, we compare three benchmark models. The first captures the original substitution view we introduced earlier: if brokers mainly help citizens overcome weak public assistance, expanding state capacity should reduce their value and lower the brokered share of applications. That prediction contradicts disproportionate expansion of brokered applications in the data. A second, more pessimistic, model allows brokers to obtain priority from approving officials using speed money but leaves application-stage productivity unchanged. Assuming approvers' time is constrained - which is why priority has a price to begin with - this framework cannot simultaneously predict increase in brokered applications and a narrowing processing time gap between direct and brokered applications. We then consider a simple complementarity model which can rationalize our joint set of empirical results: additional application-stage frontline staff

reduce the time and effort brokers spend to submit applications per client. Brokers indirectly also save frontline operators' time by bringing in ready applications, so operators save the time they spend to ensure completeness in applications (reducing the need for application rework that adds significantly to processing times in our setting). Lower broker operating costs, due to the time saved, makes broker services cheaper, and increases the number of applicants who can afford brokers, thereby expanding the broker market.

However, complementarities between brokers and frontline operators can coexist with a broker advantage later in the process. We find little evidence that *Sewa Kendra* operators themselves give brokers priority: brokered applications are no more likely than direct applications to enter the system earlier in the day. This is the main margin through which these workers could move brokers ahead. The pattern changes once applications reach approving officials. We compare approval before the statutory deadline, the service-specific legal time limit for issuing a certificate, across services that give these officials more or less discretion. After the staffing expansion, the broker advantage in timely approval shrinks substantially for lower-discretion services but remains essentially unchanged for higher-discretion services. This persistence is consistent with brokers retaining an approval-stage advantage precisely where officials have greater scope to influence outcomes. The evidence does not establish how that advantage is obtained, but it leaves room for preferential treatment, including the possibility of payments for faster approval.

Our paper contributes to two main strands of literature. Firstly, we add to the growing literature on state capacity by showing complementarities in public provision between the State and private assistance markets. The existing literature studies how government performance depends on the recruitment and quality of public employees, the organization of bureaucracies, availability of public resources, and ability to raise revenue (Dal Bó et al., 2013; Muralidharan et al., 2023; Ashraf et al., 2020; Romero and Singh, 2022; Best et al., 2023; Ganimian et al., 2024; Aneja and Xu, 2024; Barnwal et al., 2026; Aman-Rana and Minaudier, 2026). Related work shows that actors outside the formal bureaucracy can also shape state performance: Balán et al. (2022), for example, show that local chiefs improve tax collection by supplying information and local knowledge that state agents lack, even while collecting more bribes. We complement this rich but relatively new literature by carefully identifying effects of an additional frontline worker on both the number of citizens that a state is able to serve, as well as how efficiently it can do so. Our state capacity expansion occurs where there are sequential production stages (which is often the case), and expanding staffing at one stage can change both service delivery and the costs of private intermediation. Our results are consistent with complementarities between frontline workers and brokers: additional public staff can lower brokers' costs of assisting applicants even as applying directly becomes easier. Understanding these market responses can help governments plan future capacity expansions by accounting for the additional demand intermediaries may generate and the stages of the administrative process that

must accommodate it.

Second, we speak to the literature on assistance in accessing public benefits. The costs of accessing public benefits depend on government procedures and on how private assistance is supplied. A large literature shows that information, paperwork, and application assistance can substantially affect take-up of public benefits (Bettinger et al., 2012; Bhargava and Manoli, 2015; Finkelstein and Notowidigdo, 2019; Banerjee et al., 2021; Gupta, 2026). Much of this work examines assistance provided by governments, researchers, or other organizations, with lower focus on how paid assistance is supplied and its distributional effects. Byrne et al. (2025) for instance study how private participation responds to public policy: higher SNAP spending during the Great Recession, for example, led more retailers to join the program. We study a different form of private participation: intermediaries who citizens pay to obtain government documents that are needed to access social welfare schemes and other services. We document an endogenous market for assistance whose use and supply respond to public capacity: both direct and broker-mediated applications increase following the staffing expansion, alongside entry by intermediaries. Understanding who uses these private assistance markets and where assistance remains scarce or costly can help governments identify where additional public support is needed. It can also inform whether and how intermediaries might even be incorporated into formal service delivery through transparent fees and recorded transactions[3].

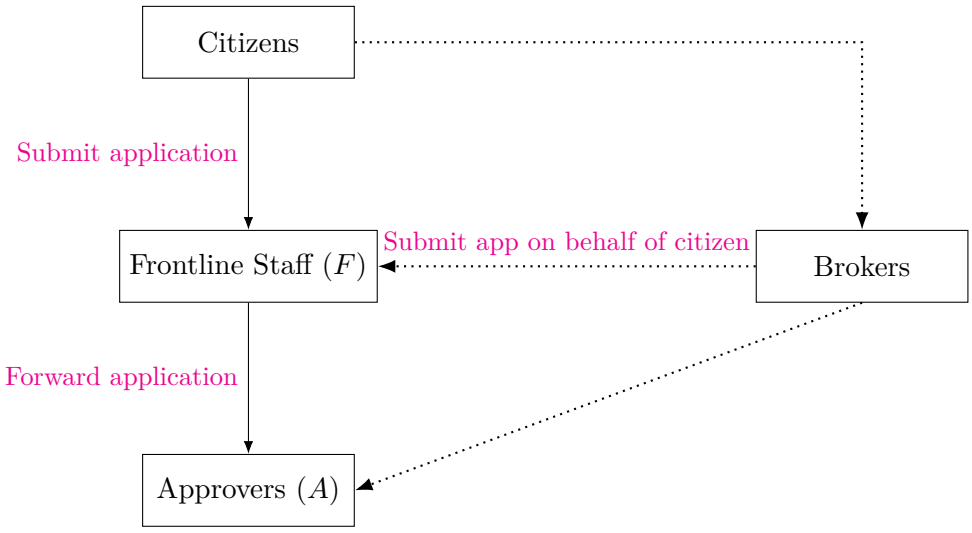
The remainder of this paper is organized as follows. Section 2 outlines the institutional background and the staffing reform, section 3 introduces the data and variables of interest, section 4 details our empirical strategy, and section 5 presents descriptive evidence on how our outcomes of interest evolved following the reform. Section 6 reports our estimation results and estimated elasticities, with Section 7 presenting the benchmark models. Section 8 discusses original mystery shopper evidence on broker price quotes to describe the market. Finally, Section 9 discusses implications for citizen welfare, and Section 10 concludes.

2. Institutional Background and Staffing Reform

In this section we describe how an application becomes a certificate, role of different staff at the two production stages (application submission and approval), and the staffing reform underlying our empirical strategy of March 2023. We use confidential administrative data from a large Indian state with a population of roughly 32 million. We consider this state to be an ideal setting to examine our question because the state has invested in a unique digital governance infrastructure called the *Sewa Kendras* (SKs).

[3]Startups such as Haqdarshak very recently have started working with private organizations and state governments by providing assistance to individuals

Figure 1. Two-stage production process



2.1. Sewa Kendras and the application process

An overview of SKs. These are service centers acting as an interface between the citizen and the state. They simplify the process of filing applications by acting as centralized kiosks for submitting applications, general queries, grievances, and miscellaneous post-office related work pertaining to all departments. Citizens can use these kiosks to access basic services like certificate issuance, record updating, and license and permit applications. There are 516 SKs in the state, each catering to roughly 45,000 citizens. First inaugurated in August 2016, there were more than 2,000 SKs before 2017 with very low footfall (providing only 170 services). After 2017, existing *Suvidha* Centers, *Saanjh Kendras*, and others were amalgamated because they were not financially remunerative, reducing the number to 516 SKs.

Services provided. Currently SKs deliver over 300 government services, of which about 30 percent are 'online services', i.e. entire process of application, examination, approval and delivery is managed electronically (and in fact no physical records or files are maintained). While citizens in theory can complete an application online, in practice nearly all citizens default to submitting their applications in-person at the SK because of recurring technical issues on the online portal. The certificates we study in this paper are a subset of the services that the SKs provide. Our analysis focuses on the subset of services for which we received complete application-level administrative records, including submission and issuance dates. These include income, caste, residence, birth and death certificates, pensions, and related administrative approvals across six government departments. Besides identity certificates, services like updating demographic and biometric information, land mutation, driving

license updates, among others are also processed by SK operators.

Types of SKs. There are three types of SKs in the state: type-I SKs are located in district headquarters with 12 counters on average; type-II SKs cater to urban population with 4 counters; and type-III SKs cater to rural population with 2 counters.

How SKs work. As soon as they arrive, citizens can get a token number through the helpdesk at the SK. They wait for their turn in the designated waiting area until their token number is displayed/announced by a SK operator. Citizens then proceed to the respective counter that called their token number and interact with the SK Operator to submit their application. After successfully submitting the application, the operator provides the citizen with a fee receipt. The official fee for many services are near zero but a few services have a maximum facilitation fee of about INR 50-70 (equivalent to USD 0.52-0.73). Using the unique application ID generated, citizens can track the status of their applications online. Operators usually forward the application to approving officials who verify eligibility and all documents and approve applications. Once the application is reviewed and approved, citizens can collect the final certificate by visiting the same SK they originally submitted the application.

SK operators. All SK operators (more than 50 percent female) hold at least an undergraduate degree. Many type-I and II operators were private sector employees before they joined, unlike operators in type-III SKs. They receive recurring software and computer training for professional development. Citizens rely on these frontline operators to assemble, fill out, and informally verify documents before they are forwarded to department officials. The operators put together an applicant's package, enters all application details in the online database and forwards the application to a higher government official who is responsible for verifying and approving the application. Figures 2 and 3 illustrate the frontline operator set-up. In March 2023, the policy reform we study is a statewide increase in their hiring.

Approving officials. After a SK operator submits an application, it is sent to the government department responsible for that service. Officials in these departments verify the applicant's eligibility and supporting documents and decide whether to approve, or send the application back to be redone, or in very rare cases (less than 0.1 percent), reject. These officials are separate from SK operators and are generally based in district, sub-district, departmental, or other government offices rather than at the service centers themselves. The organization of approval varies across services. Some departments, such as revenue, health, and policing, have extensive local offices and staff, while others

Figure 2. Citizen-facing SK counters



Figure 3. Behind SK counters with operators



rely on a smaller number of district-level or centralized officials. Approving officials also typically perform other administrative duties, so the number of officials formally assigned to a department does not necessarily measure the capacity available for processing these applications. Vacancies are common in several departments, further limiting effective approval capacity. Each service is governed by the state's Right to Services Act, which specifies the official responsible for approval and the maximum time allowed for delivering the service. Applications may be rejected, but rejections are rare in our data. Importantly for our empirical design, the frontline staffing expansion increased the number of SK operators without a corresponding increase in these downstream approving officials.

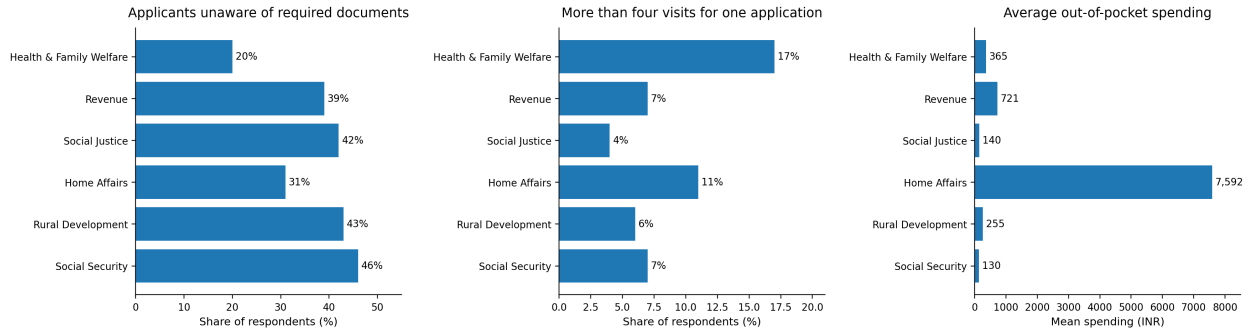
2.2. Citizens' SK experience: visits and out-of-pocket expenses

Existing survey evidence points to substantial frictions in navigating these services. Barnwal et al. (2026) surveyed applicants across six government departments and find that 44 percent did not know which documents were required before arriving at a Sewa Kendra, with department-level shares ranging from 20 to 46 percent. Some applicants also make repeated visits: 7 percent report visiting a Sewa Kendra more than four times for the same application, with the share ranging from 4 to 17 percent across departments. Out-of-pocket spending can also be substantial relative to the low official fees. Median reported expenditure is INR 180 overall, with department medians ranging from INR 40 to INR 5,000. These patterns suggest that obtaining a government certificate can involve information costs, repeated interactions with the bureaucracy, and meaningful private expenditure (Figure 4).

2.3. The staffing reform

On 10 March 2023, during the State budget, the government announced an allocation of roughly USD 7.4 million towards e-governance. They did not specifically outline how the money was to be spent. However, they did highlight their focus on improving public service delivery and filling up

Figure 4. Survey results from Barnwal et al. (2026)



vacancies. We examine staff records from the state's administrative data and find no change to approver hiring rates in the sample period around the state budget. However, we do find a rapid growth in the number of operators in the system. Our conversations with officials managing the SKs assure us that this is because hiring operators is much cheaper and less politically charged than hiring higher government officials like approvers.

On March 10, 2023, the state government implemented a large staffing expansion across administrative offices (Figure 5). Staff onboarded on the government portal roughly tripled, from about 50 employees per day before the shock to about 150 employees per day immediately after the shock, with daily new staffing reaching roughly 175–200 employees by August 2023.

Further, since operators work out of SKs, assignment of operators are correlated with the existing SK infrastructure in the state. No new SK was opened. This meant that districts with greater SK infrastructure, and therefore higher existing staff per capita, were also mechanically exposed more to the mass operator hiring at the state-level.

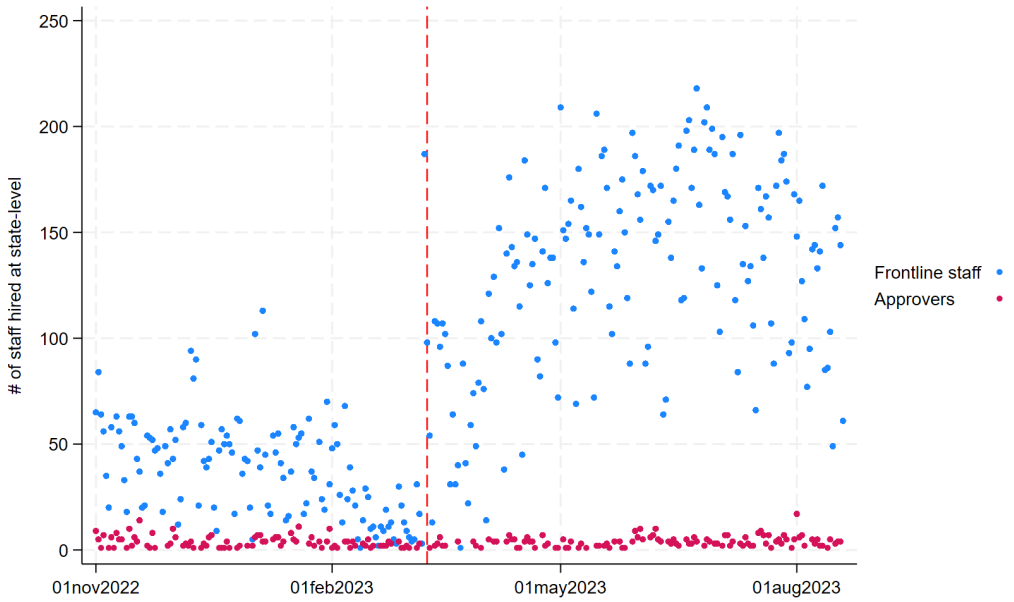
The time-series shock comes from a statewide staffing expansion during the sample period. Conceptually, we view this as the expansion of intake capacity only instead of a proportional expansion of the whole production process.

3. Data and Measurement

3.1. Administrative application records

The state began digitizing the universe of all applications and their processing times from 2018 in an effort to increase transparency and improve public service delivery. Granular administrative records on applications and issuances of certificates for this state are not routinely available to

Figure 5. Staffing expansion and first-stage variation



researchers. This was a result of sustained relationship-building and trust with the government and department officials. The resulting data in this paper cover November 2022 through mid-August 2023. Earlier records were unavailable because political developments[4] made it difficult for the department to continue sharing data. We end the analysis mid-August because statewide floods led to (i) large-scale reassignment of government staff for flood relief from that date, and (ii) staff responsible for maintaining the administrative database were also affected, creating delays in the updating of application records.

The data cover the universe of public service applications submitted through government service centers between November 1, 2022 and August 19, 2023. The sample period ends right when large floods hit the state in August 2023 leading to mass reallocation of government staff for flood relief.

Administrative data overview. The administrative data contains approximately 1.2 million applications across 24 districts and 6 government departments. The services include income certificates, caste certificates, residence certificates, pensions, birth and death certificates, etc and related administrative approvals. The administrative data record application submission dates, certificate issuance dates, service categories, district identifiers, and the mobile number associated with each application. Processing time is measured as the number of days between submission and certificate issuance. Processing times use subsequent approval dates, including approvals occurring

[4]Farmers protests where the state faced severe disruptions and large-scale unrest.

after the submission period shown on the horizontal axis. Applications submitted near the end of the plotted period therefore remain observable beyond that period. We also compare processing times to service-specific statutory deadlines to construct delay outcomes. The main analysis aggregates outcomes at the district-by-date level.

Policy variable of interest. The key policy variable of interest for us is frontline operator staffing per capita at the district-date level. Let F_{dt} denote frontline operators per 100,000 residents in district d on date t, using 2011 Census population for the denominator. The administrative staffing data distinguish frontline service operators, who are the first point of citizen contact at application centers, from approving officials with greater authority over verification or approval.

3.2. Identifying brokered applications

Citizens routinely use brokers to file applications but since they are unregulated, administrative data do not record whether an application was filed through a broker. We infer likely broker activity from the repeated use of a single mobile number across applications. Most citizens apply for a given certificate rarely: some certificates are needed once, and others must be renewed about every five years. A citizen applying on their own behalf should therefore appear on few applications during our ten-month sample. Figure 6 shows the distribution of applications per mobile number. About 90 percent of numbers appear on only one or two applications, and the distribution has a long right tail. We classify an application as broker-mediated if its mobile number appears on three or more applications during the sample. Government officials and the department's IT staff shared with us that they use the same rule to identify likely broker activity.

Misclassification of the broker proxy. Our broker measure is a proxy and can misclassify applications in either directions. It misses brokers who enter their clients' own mobile numbers. It can also count members of a household who share one number, or unpaid helpers who file for relatives and neighbors as brokers. If these errors are unrelated to application outcomes, they attenuate measured differences between brokered and direct applications. Additionally, our broker classification uses the full sample. This means a number with few applications before the reform can cross the threshold only because of applications filed afterward. To address this, we show that our results are similar when brokers are classified using pre-reform applications only, and with different thresholds for filed applications (Table 12). Another concern we consider is whether frontline operators themselves might be entering their own mobile numbers when filing for some applicants. We find this to be unlikely because our classified broker mobile numbers are observed to submit applications across multiple counters and service centers which is infeasible for a frontline

operator (Figure 22).

Figure 6. High frequency mobile numbers identify likely broker activity



Notes: Each bar reports the percentage of all applications associated with mobile numbers that appear on the indicated number of applications over the study period. The dashed vertical line marks the classification cutoff: applications associated with mobile numbers appearing on three or more applications are classified as brokered.

3.3. Outcomes and variable construction

This classification allows us to construct brokered application volume, broker share of total applications, active brokers, and the direct-broker observed processing-time gap in district d at time t. The main broker-market outcomes are:

$$\text{Broker share}_{dt} = \frac{\text{Brokered applications}_{dt}}{\text{Total applications}_{dt}},$$
$$\text{Active brokers}_{dt} = \text{Number of high-frequency phone numbers active in } (d, t)$$

The processing-time outcomes include mean processing time, the fraction of applications delayed beyond statutory deadlines, processing time for brokered applications, processing time for direct applications, and the direct-broker time gap. The time gap is defined as

$$\text{Gap}_{dt} = \overline{T}^{D}_{dt} - \overline{T}^{B}_{dt} \tag{1}$$

where $\overline{T}^{D}_{dt}$ and $\overline{T}^{B}_{dt}$ are mean processing times for direct and brokered applications.

Observed processing time gap between direct and brokered applications is the mean difference in days from application submission to a certificate being issued by each of the two channels. We aggregate this at the district x date level. Table 1 shows substantial cross-district variation in processing times in the pre-reform period. In most districts brokered applications are processed faster on average, but the magnitude varies considerably and in a few districts the difference is small.

Table 1. Pre-shock applications, broker share and processing times by district

District	Average daily applications per 100,000	Brokered applications (%)	Nonbroker days	Broker days	Nonbroker minus broker	p-value
District 1	9.92	14.91	25.10	15.18	9.91	< 0.001
District 2	10.43	16.25	11.99	11.26	0.74	0.165
District 3	11.02	18.79	17.57	12.08	5.49	< 0.001
District 4	8.67	16.74	12.11	12.89	-0.78	0.108
District 5	8.46	13.03	9.64	7.81	1.84	< 0.001
District 6	12.74	25.37	12.15	9.06	3.09	< 0.001
District 7	4.86	15.86	18.25	16.49	1.76	0.001
District 8	8.67	27.65	3.44	2.85	0.59	< 0.001
District 9	12.80	19.72	5.02	3.65	1.37	< 0.001
District 10	8.06	15.56	11.91	9.13	2.79	< 0.001
District 11	12.36	21.75	9.91	9.97	-0.06	0.843
District 12	5.61	15.94	14.63	15.55	-0.91	0.008
District 13	12.29	28.97	14.55	10.79	3.76	< 0.001
District 14	8.84	13.59	22.94	12.69	10.25	< 0.001
District 15	17.57	23.38	8.37	8.89	-0.52	0.014
District 16	8.59	15.82	22.12	18.55	3.57	< 0.001
District 17	13.01	24.14	8.00	5.57	2.44	< 0.001
District 18	9.65	17.02	13.82	12.37	1.45	0.005
District 19	14.60	20.32	18.78	12.16	6.63	< 0.001
District 20	10.89	22.77	16.77	11.32	5.45	< 0.001
District 21	11.36	18.80	17.04	9.17	7.87	< 0.001
District 22	14.06	35.28	19.77	7.21	12.56	< 0.001

Notes: We anonymize district names following out Data Use Agreement with the State Government. All columns describe applications created from Nov 2022 through 9 Mar 2023. Processing time is the time (in business days) taken from application filing to issuance. These are separated by direct and brokered applications. The difference is mean non-broker processing time minus mean broker processing time. Two-sided p-values test equality of the means using a separate application-level regression in each district, with an intercept and broker indicator and standard errors clustered by application creation date.

4. Empirical Design

4.1. OLS and the endogeneity of staffing

A simple OLS specification would estimate:

$$Y_{dt} = \beta F_{dt} + \nu_d + \tau_{dow} + \varepsilon_{dt}, \tag{2}$$

where Y_{dt} is an outcome in district d on date t, F_{dt} is frontline operators per 100,000 residents, ν_d are district fixed effects, and τ_{dow} are day-of-week fixed effects. The coefficient β measures the association between staffing and the outcome within districts, after controlling for day-of-week seasonality. However, OLS may be biased because staffing could respond to local demand, administrative backlogs, or other unobserved district-level conditions.

Realized frontline staffing may respond to local demand, administrative backlogs, or other conditions that also affect our outcomes. If we run a simple OLS regression, we may therefore confound effects of additional workers with unobserved conditions that led to their assignment. To address this concern, we instrument realized staffing with the interaction between pre-reform frontline staffing per capita and the statewide staffing expansion. This instrument uses predetermined differences in districts' exposure to the expansion rather than their realized staffing adjustments alone.

4.2. Instrument Variable

To address the OLS concerns, we use a differential exposure to a statewide staffing shock. The empirical design relies on differential district-level exposure (Figure 7) to a statewide increase in hiring of frontline operators (Figure 5) holding new hiring of approvers constant.

The instrument interacts a district's pre-shock staffing per 100,000 population with the aggregate statewide change in frontline staffing. Let $StaffPer100K_{d,pre}$ denote frontline staff per 100,000 residents in district d, averaged over Nov 2022 to 9 March 2023, using 2011 Census population. Let $\Delta AggregateFrontStaff_t$ denote the change in total frontline staff statewide from the mean pre-shock total frontline staff on day t. The instrument is:

$$Z_{dt} = \text{StaffPer100K}_{d,\text{pre}} \times \Delta\text{AggregateFrontStaff}_t. \tag{3}$$

$StaffPer100K_{d,pre}$ is fixed before the hiring expansion and does not respond to post-reform conditions in the district. $\Delta AggregateFrontStaff_t$ is common to all districts. If new staff are allocated in proportion to existing staff, Z_{dt} predicts the district's share of the statewide increase.

The first stage is

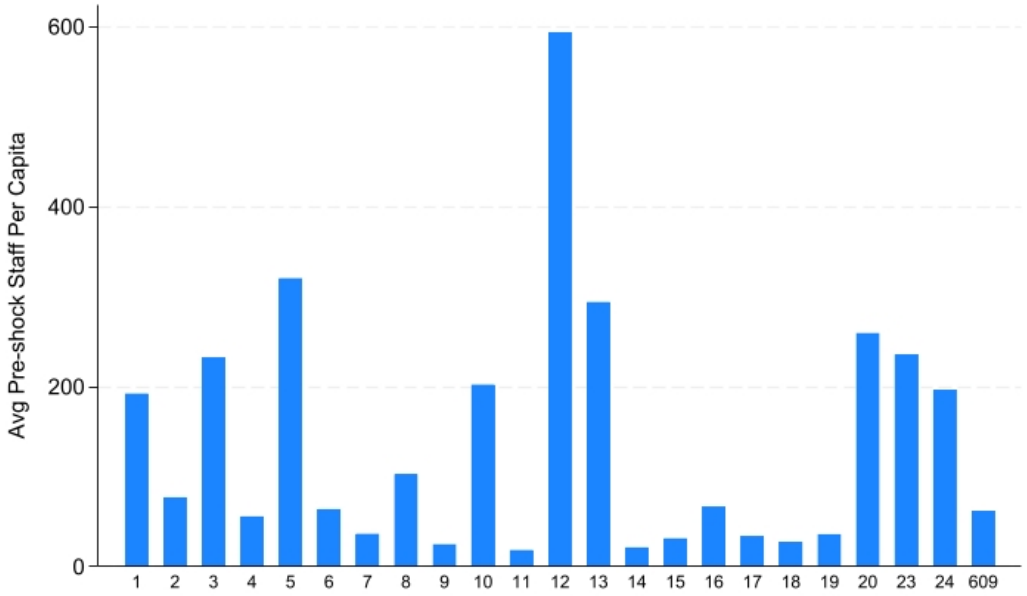
$$F_{dt} = \gamma Z_{dt} + \nu_d + \tau_{dow} + u_{dt}, \tag{4}$$

and the second stage is

$$Y_{dt} = \beta \widehat{F_{dt}} + \mu_d + \kappa_{dow} + \epsilon_{dt} \tag{5}$$

where $\widehat{F}_{dt}$ is the frontline staff per 100,000 residents in district d at time t as predicted by the instrument, ν_d and μ_d are district fixed effects, and τ_{dow} and κ_{dow} are day-of-week fixed effects. We

Figure 7. Staff per 100,000 population by district in 2021



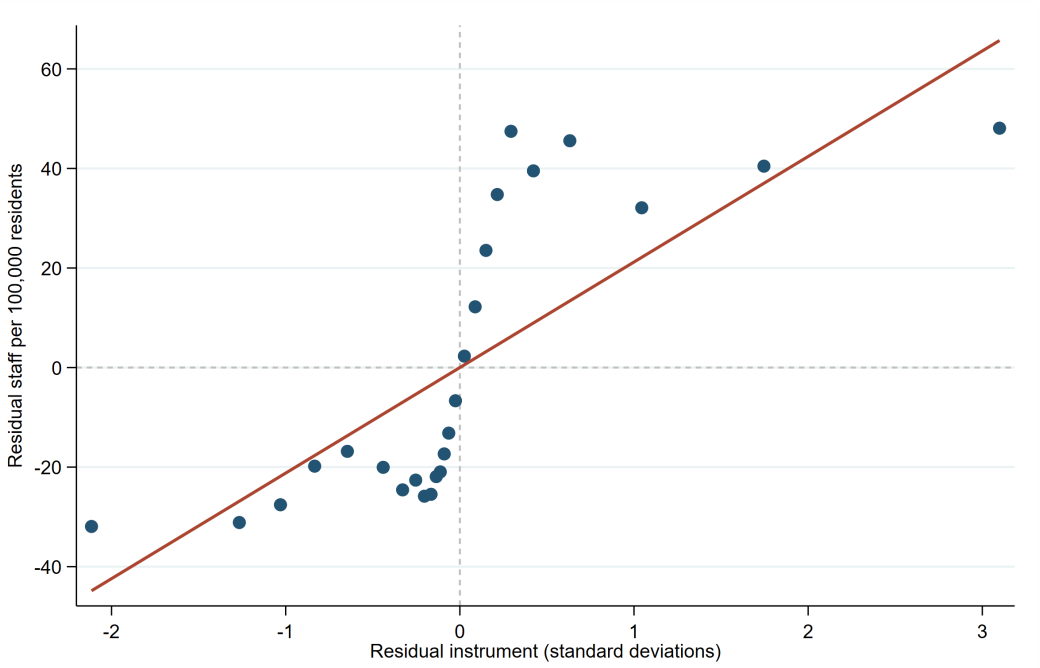
cluster standard errors at the district and date level. The first stage has an F-stat of 15.9 and is visualized in Figure 8.

Figure 8 visualizes the first stage after accounting for persistent differences across districts and regular day-of-week patterns. The horizontal axis plots the instrument after partialling out district and day-of-week fixed effects. We divide this residualized instrument by its standard deviation: a one-unit movement along the horizontal axis corresponds to a one-standard-deviation increase in predicted exposure. The vertical axis is realized frontline staffing per 100,000 residents after removing the same district and day-of-week fixed effects. Thus, a value of 20 on the vertical axis means that realized staffing is 20 workers per 100,000 above the level predicted by the district and day-of-week effects. The positive relationship indicates that greater predicted exposure is associated with higher realized frontline staffing. A one-standard-deviation increase in the residualized instrument predicts approximately 21.2 additional frontline workers per 100,000 residents.

4.3. Identification assumptions and validity

The exclusion restriction. We need staffing expansion captured by the instrument affect applications, broker activity, and processing times only through its effect on frontline staffing in the district. For example, this restriction would fail if the expansion also brought changes in application requirements, service availability, or approval procedures that independently affected these outcomes. Instrument validity also requires that, conditional on district fixed effects and the included controls, predicted staffing be unrelated to other changes affecting the outcomes. District fixed effects absorb

Figure 8. Instrument first stage



persistent differences associated with pre-reform staffing, but do not absorb changing demand or administrative conditions associated with that exposure. Measuring exposure before the reform therefore does not, by itself, establish instrument validity.

It may still be the case that operators were being targeted at specific districts based on pre-shock performance. We verify this by plotting the relationship between district-level exposure and pre-shock outcomes like applications processed and processing times in Appendix A, Figure 20. We do not find any evidence of targeting based on pre-shock outcomes. We also verify that this shock to operator hiring is not a seasonal event that occurs every year.

Our preferred specification does not include date fixed effects. We use statewide timing of the expansion as well as differences in district exposure. Including date fixed effects substantially weakens the first stage by removing the common time-series component of the instrument. To retain the time-series variation, we need to assume other statewide developments do not generate changes in our outcomes that are correlated with the staffing expansion, conditional on the included controls. We assess these concerns using institutional evidence and placebo tests.

First, we check policy margins through which a concurrent change could most plausibly affect applications or processing times. Between November 2022 and August 2023, the government did

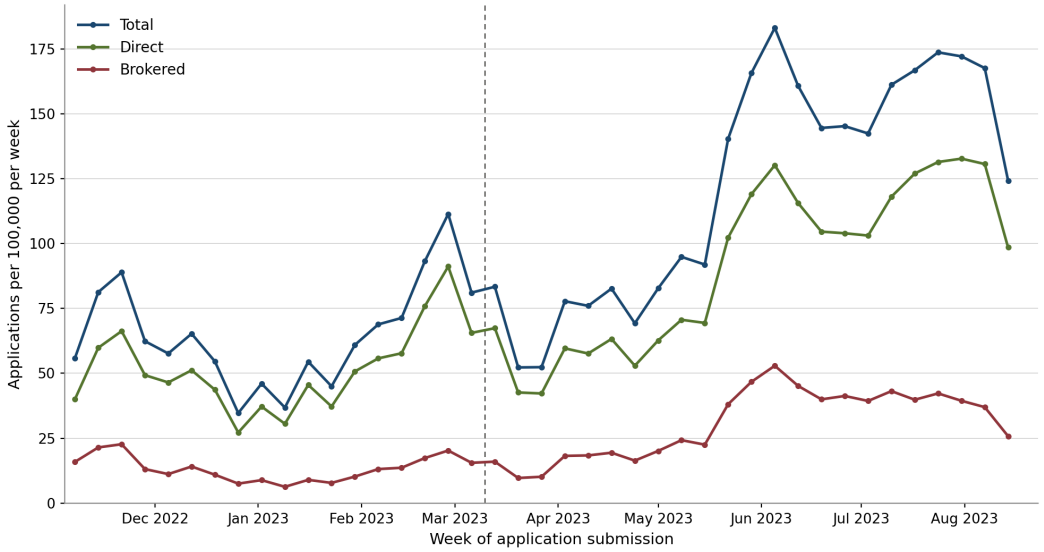
not change the fees charged for services or for facilitation at SKs, the set of services offered, or the documentation required for any service. Second, we account for seasonal events like school admissions. Demand for caste and income certificates rises seasonally during the admissions period in December to February (before classes begin in April). However, this is in our pre-shock period and if anything, the effect of seasonality would imply that applications increase before the staffing shock happens, which is the opposite in our case under descriptive statistics below. Third, we test whether the timing of the reform is plausibly exogenous. We assign placebo reform dates within the pre-period and verify that they do not affect our service delivery measures (Appendix A, Figure 21).

5. Descriptive facts around the staffing reform

5.1. Application volume and processing time

Before moving to the IV results, we show how our outcomes of interest evolved after the 10 March event. The raw time series graph plotting applications by their submission channel (direct or brokered) see a small dip around January for New Years typically being a slow time in the bureaucracy, and in end of March because of a mobile internet suspension in the state for 2 days. However, the general expansion of applications per capita is clearly visible. We show evolution of total applications segregated by application route (direct and brokered). Applications per capita at the state-level rise from roughly 60 pre-shock to about 130 after.

Figure 9. Applications per 100,000 populations

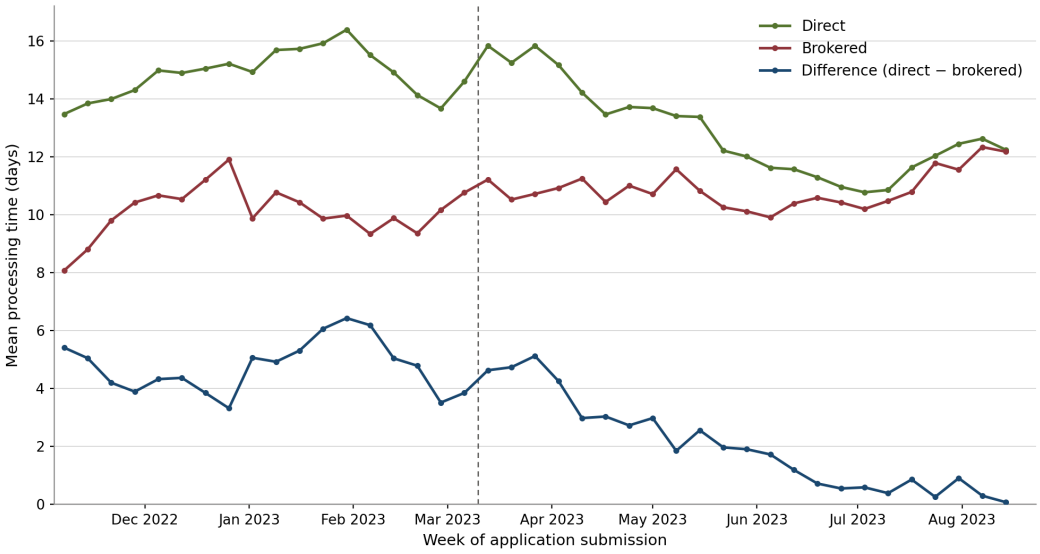


5.2. Average processing time of applications

Mean processing time, i.e. difference in business days between submission and issuance, of applications improved after the event and is driven largely by direct applications. We plot this as a raw time series trend at the state-level in Figure 10. The time taken by brokered applications remain approximately similar throughout the sample period. This is important because the observed processing time advantage that citizens may have from applying through a broker appears to diminish after the frontline operator hiring shock. Intuitively, we would expect citizens to then reduce their demand for broker services if the only service brokers offered was guaranteed faster processing after application is submitted. But that is not the case, as we see both in the share of brokered applications above in Figure 9 as well as the number of active brokers per capita across the state, as is shown in Figure 11.

Figure 10 reports mean processing times by the week in which applications were submitted, separately for direct and brokered applications. The figure shows a gradual narrowing of the processing-time gap following the staffing expansion, largely reflecting a decline in processing times for direct applications.

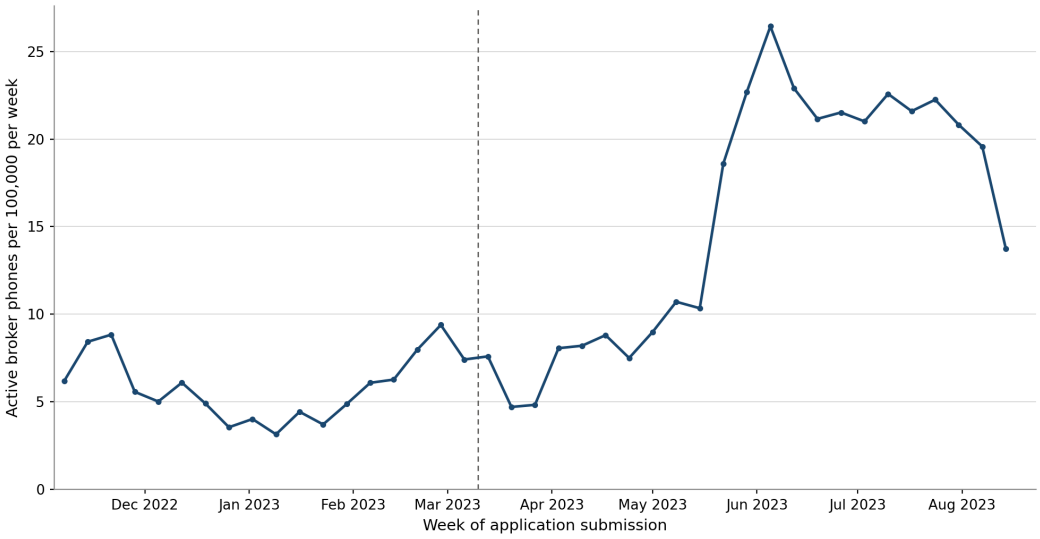
Figure 10. Processing time by application channel



6. Instrument Variable Regression Results

Throughout this section, we discuss magnitudes for an increase of one-standard-deviation increase in the instrument (i.e. for 21.2 additional frontline operators per 100,000 individuals). To get a benchmark of what these magnitudes might mean, after presenting all results, we will explicitly

Figure 11. Active brokers per 100,000 populations



calculate the magnitudes for an sample district with 600,000 population - this also happens to be the size of the smallest district in our sample.

6.1. Public service delivery outcomes

The first result is that the frontline expansion increased application volume. Table 2 reports positive and statistically significant effects on applications per capita. The IV estimate implies that applications per 100,000 residents increases by 4 relative to a pre-shock mean of 11.45. The OLS estimates are slightly smaller than our IV estimates. It makes intuitive sense for the OLS estimates to be biased towards zero since actual local staffing may have been targeted at districts. On the other hand, we also do not expect the estimates to differ by a large margin either because frontline operators make up only a part of the production process that determines application volume. It is also the stage which does not directly impact throughput since certificate issuance authority lies with approvers.

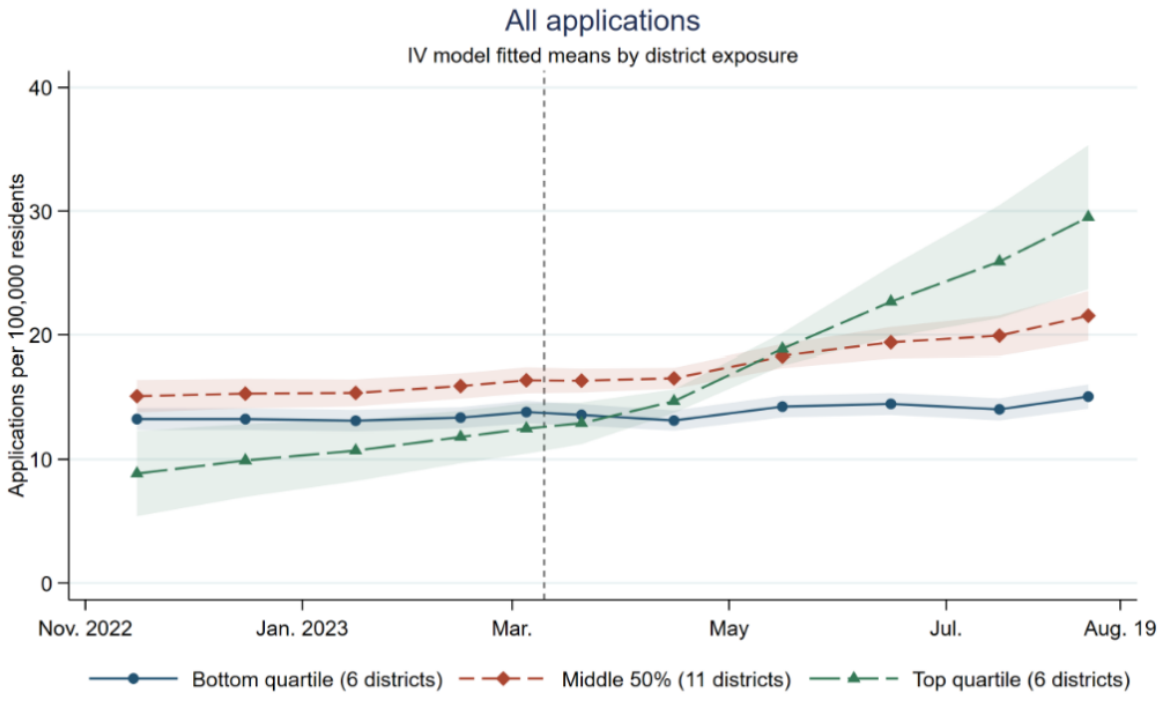
The IV estimate for average processing time of applications shows a decline of approximately 1 day relative to a pre-shock mean of 14.22 days. This is informative because the increase in applications could mechanically create congestion. The fact that average processing time declines while volume rises implies that the direct capacity effect of additional frontline operators dominates any slowdown that may have happened due to higher applications being submitted in response to the additional frontline operators.

Table 2. Effects of frontline staffing on application volume and processing time

	Applications per capita		Processing time	
	OLS	IV	OLS	IV
Frontline staff per 100k	0.154***	0.188***	−0.035**	−0.043**
	(0.019)	(0.025)	(0.011)	(0.013)
Observations	6,569	6,569	6,569	6,569
Pre-shock mean	11.45	11.45	14.22	14.22
Kleibergen-Paap F-stat		15.55		15.55
District FE	Yes	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses. Staff per capita is measured as frontline staff per 100,000 residents. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 12. Predicted application volumes by district exposure



Applications by submission channel. Table 3 reports the staffing response separately for brokered and direct applications. The IV coefficients imply that brokered applications per 100,000 residents rose by about 1,2 while direct applications rose by about 3.09 per 100,000 residents. Both estimates are positive and statistically significant.

Figure 13. Predicted processing time saved by district exposure

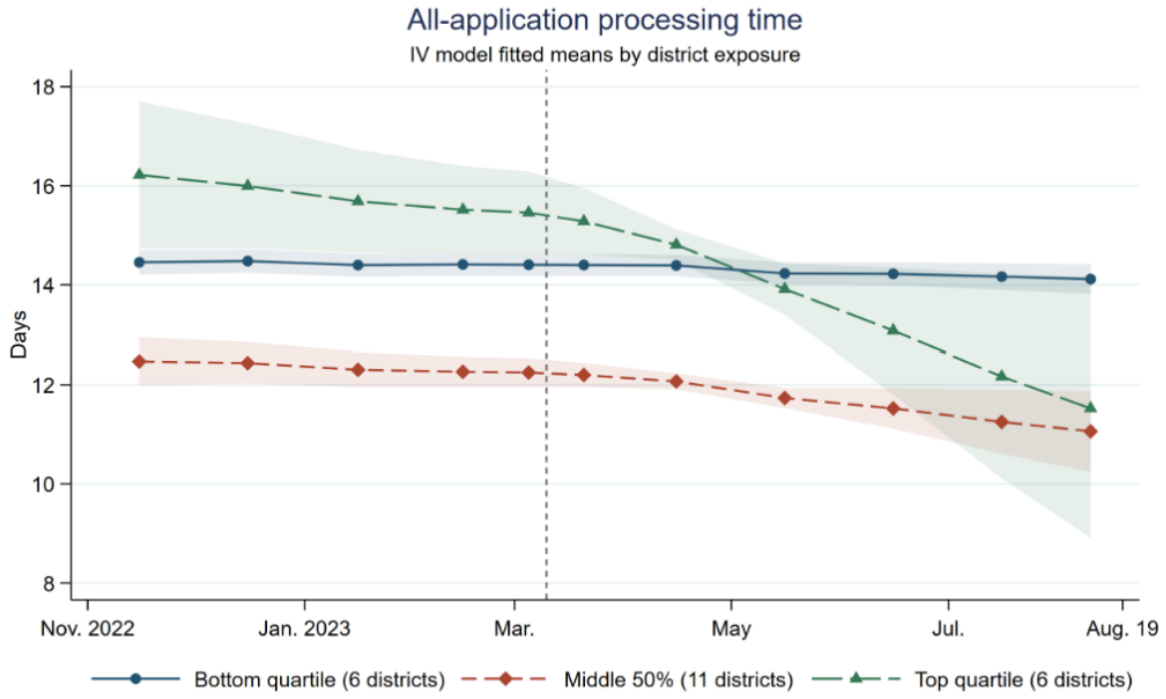


Table 3. Effects of frontline staffing on brokered and direct applications

	Brokered applications		Direct applications	
	OLS	IV	OLS	IV
Frontline staff per 100k	0.046***	0.055***	0.116***	0.146***
	(0.007)	(0.009)	(0.014)	(0.018)
Observations	6,408	6,408	6,408	6,408
Pre-shock mean	2.282	2.282	8.693	8.693
Kleibergen–Paap F-stat		15.933		15.933
District FE	Yes	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses. Frontline staffing and application volumes are measured per 100,000 residents. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

6.2. Broker market outcomes

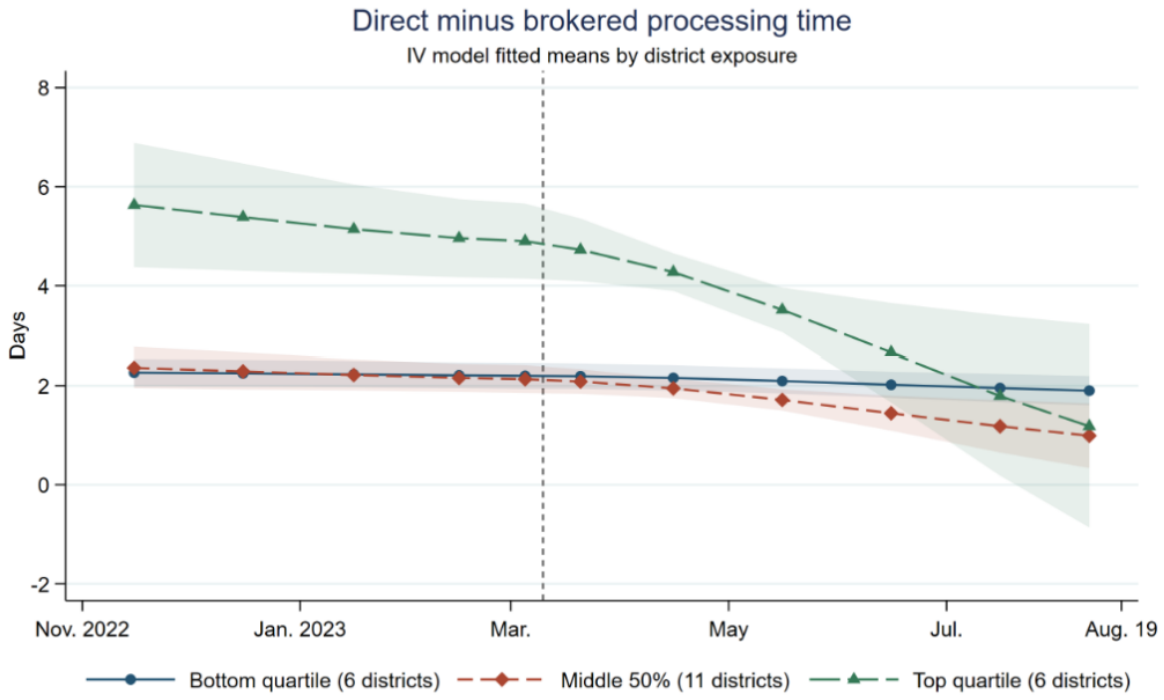
Mean processing time. Table 4 shows that the difference in mean processing time of direct and brokered applications narrowed. The IV estimate implies that the observed broker speed advantage reduced by about 0.87 days. This is important for the model because the convergence is not primarily driven by brokered applications becoming slower. Brokered processing time falls by an imprecise 0.21 days, while non-brokered processing time falls by 1.08 days. It is the direct application processing time that catches up.

Table 4. Broker speed differential and decomposition

	Speed differential IV	Brokered time IV	Non-brokered time IV
Frontline staff per 100k	−0.041***	−0.010	−0.051**
	(0.009)	(0.008)	(0.014)
Observations	6,438	6,439	6,568
Pre-shock mean	3.54	11.35	14.95
Kleibergen-Paap F-stat	15.55	15.55	15.55
District FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes

Notes: Standard errors in parentheses. Staff per capita is measured as frontline staff per 100,000 residents. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 14. Predicted speed gap for direct & brokered routes by district exposure



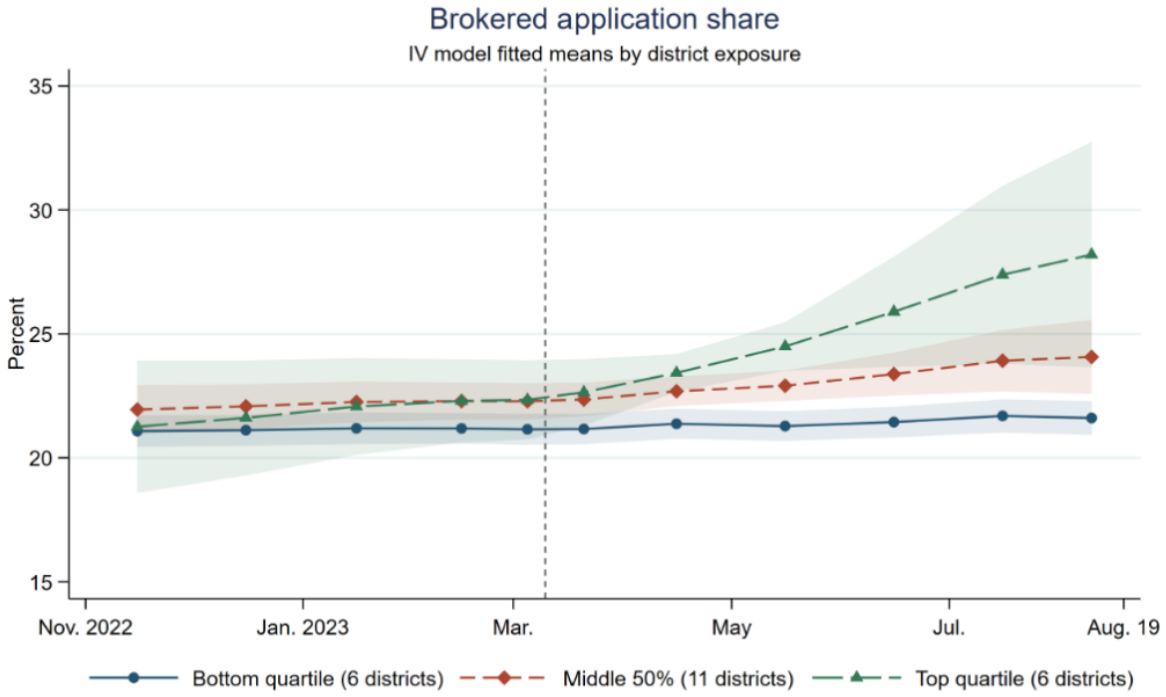
Broker market size. Despite the smaller speed gap, the broker market expanded. Table 5 shows increases in the share of total applications submitted via brokers as well as the total number of active brokers, suggesting new broker entry. We find broker share of applications rises by 1.4 percentage points relative to a pre-shock mean of 19.65. Active brokers rise by 2 relative to a pre-shock mean of 10.84.

Table 5. Broker market responses

	Broker share IV	Active brokers IV
Frontline staff per 100k	0.065***	0.092***
	(0.019)	(0.010)
Observations	6,408	6,546
Pre-shock mean	19.65	10.84
Kleibergen-Paap F-stat	15.93	15.45
District FE	Yes	Yes
Day-of-week FE	Yes	Yes

Notes: Standard errors in parentheses. Active brokers are measured per 100,000 residents. Staff per capita is measured as frontline staff per 100,000 residents. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 15. Predicted brokered share of applications by district exposure



6.3. Elasticities and economic interpretation

The pre-reform mean of frontline staffing in the main tables is

$$\bar{F}_{\text{pre}} = 75$$

workers per 100,000 population. Since the main regressions are estimated in levels, we convert each estimated coefficient into an elasticity using

$$\varepsilon_{Y,F} = \widehat{\beta}_Y \frac{\bar{F}_{\text{pre}}}{\bar{Y}_{\text{pre}}}$$

Elasticities for our outcomes of interest are outlined below in Table 6[5].

Table 6. Elasticities evaluated at pre-reform means

Outcome	IV coefficient	Pre-reform mean	Elasticity
Total applications	0.188	11.45	1.23
Broker share	0.065	19.65	0.25
Active brokers per 100 applications	0.092	10.84	0.64
Mean processing time	-0.043	14.22	-0.23
Brokered processing time	-0.010	11.35	-0.07
Direct processing time	-0.051	14.95	-0.26
Direct-minus-broker time gap	-0.041	3.54	-0.87
Application volumes estimated separately			
Brokered applications	0.055	2.282	1.70
Direct applications	0.146	8.693	1.09

Brokered share of marginal applications. The channel-specific IV coefficients also show how the additional applications are divided between the two routes. Let $\widehat{\beta}_B$ and $\widehat{\beta}_D$ denote the staffing coefficients for brokered and direct applications. Since total applications equal the sum of the two channels, using the same specification, sample, and weights gives an implied total response of $\widehat{\beta}_B + \widehat{\beta}_D$. The brokered share of marginal applications is therefore

$$\widehat{m}_B = \frac{\widehat{\beta}_B}{\widehat{\beta}_B + \widehat{\beta}_D} = \frac{0.055}{0.055 + 0.146} \simeq 0.274.$$

Thus, about 27.4 percent of the additional applications implied by these estimates are brokered. The corresponding baseline share is the ratio of pre-reform mean brokered applications to mean total applications in the same sample:

$$\bar{s}_{\text{pre}} = \frac{\bar{Q}_{B,\text{pre}}}{\bar{Q}_{B,\text{pre}} + \bar{Q}_{D,\text{pre}}} = \frac{2.282}{2.282 + 8.693} \simeq 0.208.$$

Brokered applications therefore account for about 20.8 percent of baseline application volume, compared with 27.4 percent of marginal applications. The difference is approximately 6.6 percentage

[5]For comparison, the accounting identities $Q_B = S_B Q$ and $Q_D = (1 - S_B)Q$ imply $\varepsilon_{Q_B,F} = \varepsilon_{Q,F} + \varepsilon_{S_B,F}$ and $\varepsilon_{Q_D,F} = \varepsilon_{Q,F} - [S_B/(1 - S_B)]\varepsilon_{S_B,F}$. Applying these expressions to the total-application elasticity of 1.23, broker-share elasticity of 0.25, and broker share of 0.1965 gives 1.48 and 1.17, respectively.

points, or a 31.6 percent higher share relative to baseline:

$$100\left(\frac{\widehat{m}_B}{\bar{s}_{\text{pre}}} - 1\right) \simeq 31.6\%.$$

6.4. Interpreting the estimated magnitudes in a district of 600,000 residents

Here we translate the IV estimates into additional applications, changes in processing time, and changes in broker use. We use a district with 600,000 residents, approximately the population of the smallest district in the sample, and a staffing increase of 21.2 frontline operators per 100,000 residents. This is the first-stage response to a one-standard-deviation increase in the residualized instrument. At this benchmark, the estimates imply about 24 additional applications per day and a decline of about 0.91 business days in mean processing time. Applied to a baseline day's application volume, the time reduction amounts to approximately 63 application-business-days of waiting saved.

6.4.1. Define the staffing increase

The first-stage figure removes district and day-of-week fixed effects from the instrument and from realized frontline staffing. It then divides the residualized instrument by its standard deviation. A one-unit increase in this standardized measure is therefore a one-standard-deviation increase in exposure after removing those fixed effects. The fitted slope in the figure is approximately 21.2 workers per 100,000 residents.

Equivalently, let $\widetilde{Z}$ denote the residualized instrument, $s_{\widetilde{Z}}$ its standard deviation, and $\widehat{\gamma}$ the first-stage coefficient on the unstandardized instrument. The staffing benchmark is

$$\Delta F = \widehat{\gamma}\, s_{\widetilde{Z}} \simeq 21.2 \quad \text{workers per 100,000 residents.}$$

In the illustrative district, the equivalent increase in the number of frontline workers is

$$\Delta N_F = m\, \Delta F = 6 \times 21.2 = 127.2 \quad \text{workers in the district.}$$

This is an increase in the staffing level.

Translating coefficients. Each IV coefficient measures the response to one additional frontline worker per 100,000 residents. For an outcome Y, the change in its original regression units is

$$\Delta Y = \widehat{\beta}_Y \times 21.2.$$

If Y is a count expressed per 100,000 residents, multiply this result by six to obtain the district count. If Y is measured in days or percentage points, retain those units without multiplying by six.

The shortcut $\widehat{\beta}_Y \times 127.2$ produces a district count only when the outcome is itself a count measured per 100,000 residents.

6.4.2. Calculate the additional daily applications

The IV coefficient for total daily applications is 0.188. Its units are additional applications per day per 100,000 residents, for each additional frontline worker per 100,000 residents. For the staffing increase we have that:

$$\Delta q = 0.188 \times 21.2 = 3.9856 \quad \text{applications per day per 100,000 residents.}$$

Converting the application rate into a district count:

$$\Delta Q = 3.9856 \times 6 = 23.9136 \simeq 24 \quad \text{additional applications per day.}$$

These are additional submissions entering the system.

The pre-reform mean is 11.45 daily applications per 100,000 residents. Using this sample mean for the sample smallest district gives us:

$$Q_0 = 11.45 \times 6 = 68.7 \quad \text{applications in a baseline day's cohort.}$$

The predicted increase is approximately 34.8 percent of baseline volume:

$$100 \times \frac{3.9856}{11.45} = 100 \times \frac{23.9136}{68.7} \simeq 34.8\%.$$

The daily volume rises from 68.7 to approximately 92.6 applications under the same linear calculation.

6.4.3. Calculate changes in processing time

Processing time measures elapsed business days between submission and issuance. Its coefficient is already in days per application, so the population conversion is unnecessary. The mean-processing-time coefficient is -0.043:

$$\Delta T = -0.043 \times 21.2 = -0.9116 \quad \text{business days per application.}$$

Mean recorded processing time therefore falls by approximately 0.91 business days. Relative to the pre-reform mean of 14.22 business days, this is a reduction of about 6.4 percent:

$$100 \times \frac{-0.9116}{14.22} \simeq -6.4\%.$$

For the two application channels and their time gap,

$$\Delta T_D = -0.051 \times 21.2 = -1.0812,$$
$$\Delta T_B = -0.010 \times 21.2 = -0.2120,$$
$$\Delta(T_D - T_B) = -0.041 \times 21.2 = -0.8692.$$

Direct applications become about 1.08 business days faster. The brokered-time point estimate is a reduction of about 0.21 business days and is not statistically significant. The direct-minus-broker time gap narrows by about 0.87 business days.

6.4.4. Calculate the change in broker share

Broker share is measured on a 0–100 percentage scale. The coefficient of 0.065 means a 0.065-percentage-point increase for each additional frontline worker per 100,000 residents. At the staffing benchmark,

$$\Delta s_B = 0.065 \times 21.2 = 1.378 \quad \text{percentage points.}$$

The predicted change is approximately 1.38 percentage points. On a 0–1 scale, the same calculation is $0.00065 \times 21.2 = 0.01378$.

Starting from the pre-reform mean share of 19.65 percent gives an illustrative share of $19.65 + 1.378 = 21.028$ percent. The increase of 1.38 percentage points is about a 7% increase relative to baseline.

We can now add up the reduction in processing time across a cohort containing a baseline day's applications. This requires two inputs calculated above:

$$Q_0 = 68.7 \quad \text{applications,} \qquad -\Delta T = 0.9116 \quad \text{business days saved per application.}$$

Their product is

$$\begin{aligned} S_0 &= Q_0 \, (-\Delta T) \\ &= (11.45 \times 6) \times (0.043 \times 21.2) \\ &= 68.7 \times 0.9116 \\ &= 62.62692 \simeq 63. \end{aligned}$$

The unit is *application-business-days of waiting saved across one baseline day's cohort.* Ten applications each finishing one business day earlier would sum to ten application-business-days saved. Here, approximately 69 applications have a mean wait about 0.91 business days shorter.

This calculation holds the baseline cohort's size and composition fixed and applies the estimated mean time reduction to that reference cohort. It uses the sample-average application rate to

Table 7. Magnitude calculations for a district of 600,000 residents

Outcome	IV coefficient	Calculation and predicted change
Total daily applications	0.188	$0.188 \times 21.2 \times 6 = 23.9136$: about 24 additional district applications per day.
Mean processing time	−0.043	$-0.043 \times 21.2 = -0.9116$: about 0.91 fewer business days per application.
Direct processing time	−0.051	$-0.051 \times 21.2 = -1.0812$: about 1.08 fewer business days.
Brokered processing time	−0.010	$-0.010 \times 21.2 = -0.2120$: about 0.21 fewer business days; not statistically significant.
Direct-minus-broker time gap	−0.041	$-0.041 \times 21.2 = -0.8692$: a narrowing of about 0.87 business days.
Broker share	0.065	$0.065 \times 21.2 = 1.378$: an increase of about 1.38 percentage points.

Notes: Calculations use rounded IV coefficients and the first-stage benchmark. Staffing is measured per 100,000 residents. The application outcome is a daily rate per 100,000 residents before conversion to the district count; processing times are in business days and broker share is in percentage points. The approximately 63 application-business-days of waiting saved are calculated separately as $11.45 \times 6 \times 0.043 \times 21.2$, using the baseline application rate.

construct the district.

Thus at the baseline application rate, the estimated reduction in processing time saves about 63 application-business-days of waiting across one day's applications in a district of 600,000 residents.

6.4.5. Collecting all magnitudes

Table 7 reports the six outcomes shown in the presentation's magnitude table. Each uses the same staffing increase of 21.2 operators per 100,000 residents. Only the district application count receives the additional population multiplier.

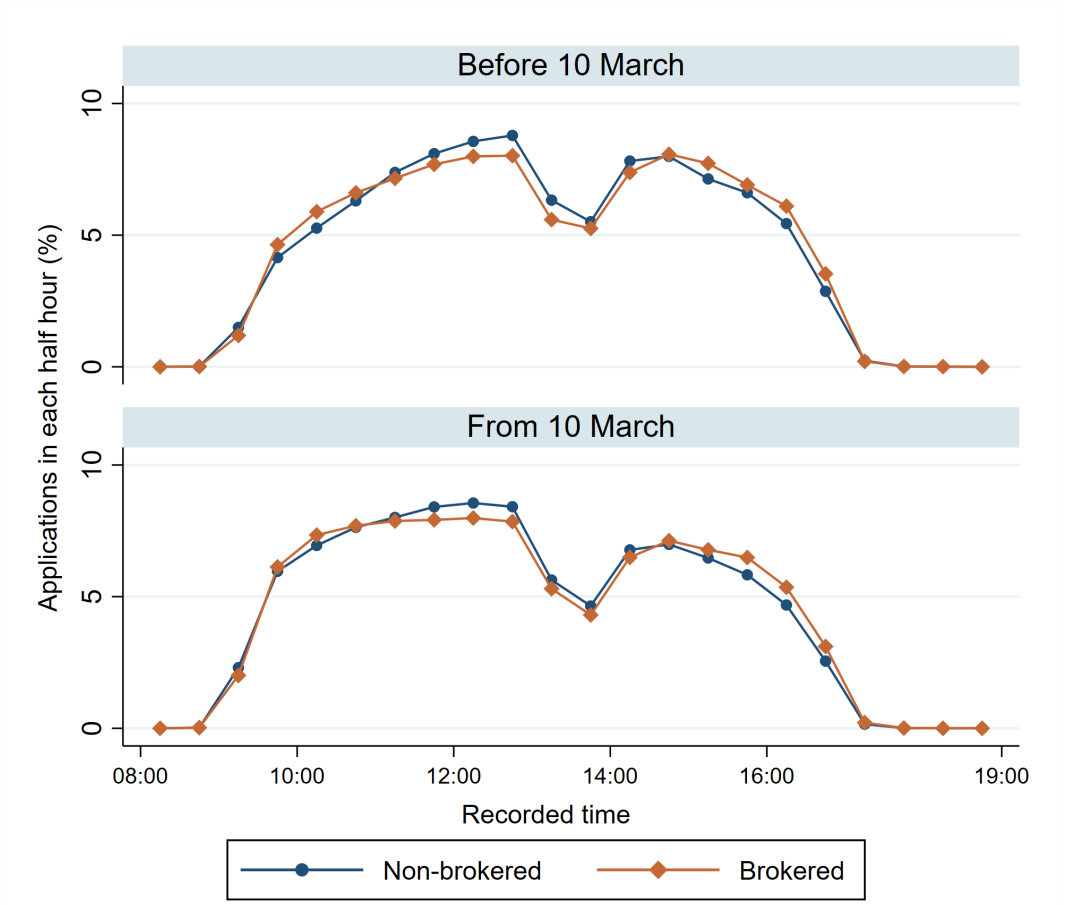
6.5. Where does the remaining broker advantage sit?

Finally we try to understand whether some other observed broker advantage persists in the system despite the converging mean processing time gap. We consider the idea of early approvals, i.e., probability of getting approved before the statutory deadline. In theory frontline operators can do this by reserving earlier tokens in the SK counters for brokers and their clients[6]. This advantage can be provided by frontline operators if they systematically let brokers submit applications early. This is the only margin on which operators can ensure brokered applicantions enter the system before direct applications and are therefore processed earlier than direct ones. We show that this is unlikely to be true in Figure 16. Both brokered and direct applications enter the system roughly at the same time throughout the day. This means that tokens for morning slots were unlikely to be

[6]During fieldwork we found this to be sometimes common although not widespread.

distributed systematically to brokers.

Figure 16. Time of application submission



Another way to obtain early approvals is when applications reach approvers. We do not observe variation at the approver stage. We also do not observe when an application reaches the approvers after being entered into the system by a frontline operator. Therefore this exercise is purely suggestive. We distinguish between discretionary and routine services based on their application requirements. Discretionary services require at least one signature from a senior government official, often the approver, whereas routine services typically rely on existing documents, such as proof of address or income, without requiring additional authorization from a senior official. Help with paperwork and submission could benefit applicants in both groups. Privileged access to approvers, however, could provide an additional advantage in services where officials have greater discretion over approval. We compare how the broker–direct gap in the probability of approval by the statutory deadline changes across these two groups following the frontline expansion. A broker advantage that persists in discretionary services but diminishes in routine services would be consistent with privileged access.

However, we cannot rule out differences in documentation requirements or applicant composition can also generate this pattern. In our sample, most health department services and income certificates are routine, whereas caste certificates, non-availability of birth/death certificates, widow pension certificates are discretionary.

For application i in service s, let r_i denote the approval date relative to that service's statutory deadline and define

$$\text{Early}_i \equiv \mathbf{1}\{r_i \leq 0\}. \tag{6}$$

Thus, Early_i equals one when the application is approved on or before its service-specific statutory deadline. Let B_i indicate a broker-associated application, let H_s indicate a high-discretion service, and let Post_t indicate the period after the frontline staffing expansion. For discretion group $g \in \{L, H\}$ and period $p \in \{\text{pre}, \text{post}\}$, define the broker early-approval gap as

$$G_{g,p} \equiv \Pr(\text{Early} = 1 \mid B = 1, g, p) - \Pr(\text{Early} = 1 \mid B = 0, g, p). \tag{7}$$

Figure 17 reports the four gaps taking into account service composition. Before the reform, broker advantage is similar across the two groups: $G_{H,\text{pre}} = 2.16$ percentage points and $G_{L,\text{pre}} = 2.43$ percentage points. After the reform, the low-discretion gap falls to 0.97 percentage points, while the high-discretion gap remains essentially unchanged at 2.17 percentage points. The low-discretion broker advantage therefore falls by about 1.46 percentage points, whereas the high-discretion advantage changes by only 0.01 percentage points. However, this is imprecisely estimated and we only consider the estimates as being suggestive.

Table 8. Decomposing the broker early-approval advantage

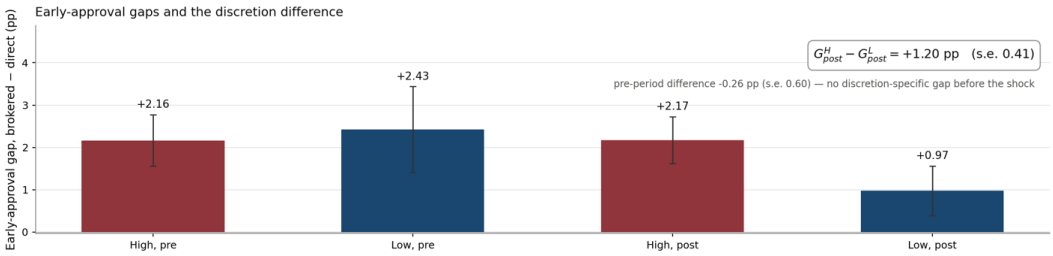
	Pre-reform	Post-reform	Change
Low-discretion services	2.43	0.97	−1.46
High-discretion services	2.16	2.17	+0.01
High minus low	−0.26	+1.20	**+1.47**

Notes: We report the percentage-point difference in the probability of early approval between brokered and direct applications. The final column is the post-minus-pre change. The lower-right entry is the DDD estimand from equation (8).

The primary estimand is the change in the high-minus-low broker gap after the reform:

$$\begin{aligned}\Delta^{DDD} &\equiv \Big(G_{H,\text{post}} - G_{H,\text{pre}}\Big) - \Big(G_{L,\text{post}} - G_{L,\text{pre}}\Big) \\ &\approx (2.17 - 2.16) - (0.97 - 2.43) = 1.47 \text{ percentage points.}\end{aligned} \tag{8}$$

Figure 17. Broker early-approval advantage by service discretion



We estimate a linear-probability specification by:

$$\begin{aligned}\text{Early}_{isdt} = \alpha_s + \mu_d + \lambda_t + \beta_B B_i + \beta_{BP}\Big(B_i \times \text{Post}_t\Big) + \beta_{BH}\Big(B_i \times H_s\Big) \\ + \beta_{HP}\Big(H_s \times \text{Post}_t\Big) + \beta_{DDD}\Big(B_i \times H_s \times \text{Post}_t\Big) + X_i'\gamma + \varepsilon_{isdt},\end{aligned} \tag{9}$$

where α_s, μ_d, and λ_t are service, district, and submission-time fixed effects. The interaction $H_s \times \text{Post}_t$ absorbs differential post-reform changes in high-discretion services that are common to direct and brokered applications. In this specification, β_{DDD} is the DDD estimand in (8).

Interpretation. The post-reform high-minus-low difference is

$$G_{H,\text{post}} - G_{L,\text{post}} = 2.17 - 0.97 = 1.20 \text{ percentage points.} \tag{10}$$

If low-discretion services provide a valid benchmark for the generic broker advantage, including applicant selection, application preparation, and other route differences that do not depend on approval discretion, then the 0.97-point low-discretion gap is a residual component and the additional 1.20 points in high-discretion services are a discretion-linked excess advantage. On this descriptive decomposition, roughly 55 percent of the post-reform high-discretion broker gap is associated with the discretion-linked component.

Besides the imprecision, we also avoid causal interpretation because high- and low-discretion services may differ along other margins. Instead, we consider this result as helping us locate a potential residual broker advantage. That is, after frontline capacity expands, the remaining broker advantage may sit in early approval concentrated in services where downstream officials retain greater discretion.

7. Benchmark models of the broker market

The empirical results present a simple puzzle. Additional application-stage workers increase applications through both channels, with a larger response through brokers. More brokers also become

active. Yet direct applications become faster, reducing the observed speed advantage of brokerage. We use three benchmarks to ask which services can account for these responses. Under the restrictions below, pure substitution predicts a falling broker share while pure approval priority predicts a widening observed time advantage. Complementarity allows brokerage to expand despite a smaller time advantage.

Pure substitution benchmark. The common belief of brokers substituting for scarce public capacity is our first benchmark model. While we show a complete proof below, we can also intuitively see how a pure substitution framework will not be able to predict all empirical facts jointly. Most importantly, if the government becomes easier to access, and there is an increase in total applications, the number of brokered applications might rise. But this model does not predict a rise in the *share* of brokered applications coinciding with falling observed speed advantage. Admittedly, citizens do not only care about the time taken after an application is submitted, they also care about the time taken to reach submission. The predictions would be affected if time taken to reach submission (which is unobserved in our administrative data) varies systematically by brokers and direct applicants. Here we rely on our observation that the time of submission does not vary greatly across direct and brokered applicants. This means that on average, brokered applications do not reach the counters earlier than direct applicants over the day. But there is a possibility of brokers having greater expertise and can generally take less time to prepare an application package – we would not be able to observe this in our data[7]. For the pure substitution benchmark, we assume away any additional advantage brokers might have that frontline workers do not, and hence they are pure substitutes.

Pure approval-stage priority benchmark. Under a framework of pure priority at approver-stage, the main job a broker has is precisely to be ahead of a direct application. This means even if total applications rise, the brokered queue becomes longer, and the value to jumping queue increases, brokers only survive as long as their speed advantage remains. This model cannot jointly predict a rise in broker share along with a falling broker speed advantage.

Complementarities benchmark. Lastly, we consider how brokers might actually work with frontline operators (since they too need to submit the application at the SK). Higher number of frontline workers can make brokers more productive by reducing the effort cost a broker incurs to reach an operator. This scenario is realistic and is also capable of jointly rationalizing all our empirical facts.

[7]We are collecting mystery shopper survey data on prices charged by brokers and the nature of services they provide with documentation. 30 brokers have been surveyed so far. We will update the draft with results as soon as we are done.

7.1. A common environment

Consider one district and one service. A citizen receives value $V > 0$ from obtaining the service and has a time cost θ. Potential applicants have constant density $\lambda > 0$ on $[0, \bar{\theta}]$. Citizens can apply directly, hire a broker at fee P, or not apply. Let Q_D and Q_B denote the flows of direct and brokered applications, $Q = Q_D + Q_B$ their total, and $S_B = Q_B/Q$ the brokered share. The value of the service and the distribution of citizens are held fixed when staffing changes.

Applications require frontline work and approval. The reform increases frontline staff-time F while holding approval capacity μ fixed. Feasible stationary flows satisfy $L = \phi_D Q_D + \phi_B Q_B < F$ and $Q < \mu$, where ϕ_j is public frontline work per application. Brokered clients' use of public workers is included in L.

Citizens care about the full application process, while the records begin at formal submission. Write R_j for preparation time and T_j for recorded processing time, where $j \in \{D, B\}$. Define

$$\tau = R_B + T_B, \qquad H = (R_D + T_D) - (R_B + T_B) > 0, \qquad \Delta = T_D - T_B. \tag{11}$$

Here τ is the full broker-route time, H is the full time advantage of brokerage, and Δ is its recorded advantage. A fall in Δ does not establish how preparation time or H changes.

The utilities of direct application, brokerage, and nonparticipation are

$$U_D = V - \theta(\tau + H), \qquad U_B = V - P - \theta\tau, \qquad U_0 = 0. \tag{12}$$

In an interior allocation (i.e. one where a broker market exists), citizens with low time costs apply directly, those with intermediate time costs hire brokers, and those with the highest time costs do not apply. The resulting demands are

$$Q_D = \lambda \frac{P}{H}, \qquad Q_B = \lambda \left(\frac{V - P}{\tau} - \frac{P}{H} \right), \qquad Q = \lambda \frac{V - P}{\tau}. \tag{13}$$

Participation therefore responds to both application time and broker fees.

Brokers have marginal cost c and fixed operating cost $K > 0$. Quantity competition and free entry determine fees, applications per broker, and active brokerage. Appendix C.1 derives this equilibrium. All comparative statics include congestion from changes in participation and route choice.

7.2. Substitution: public assistance replaces broker assistance

In this benchmark, brokers help prepare and correct applications but receive no approval priority. Their work is identical to frontline operators. They only need to additionally engage with frontline staff to submit the application package. Direct applications require more frontline handling. Increasing frontline staff leading to faster frontline interactions thus narrow the recorded gap. Brokers' marginal and operating costs remain fixed in this benchmark. Two demand forces compete: a smaller full broker advantage encourages direct application, while a shorter common application process encourages participation. A substitution prediction must account for both forces, as well as the brokers' endogenous fee.

Let $x = -d\log\tau / d\log F$ denote the proportional improvement in the full broker route and $y = -d\log H / d\log F$ the proportional reduction in its full advantage. The crowd-out benchmark maintains $y > 0$ and restricts common access gains to satisfy

$$\frac{x}{y} \leq \frac{\tau}{2\tau + H}. \tag{14}$$

Prediction 1. With constant broker costs, an interior equilibrium, and (14), additional frontline capacity reduces the brokered share. This benchmark cannot explain the observed increase in S_B.

Public assistance reduces the reason to hire a broker faster than easier access creates new demand for brokerage. Fees and participation adjust but do not reverse crowd-out under the bound. This restriction concerns full application time: the recorded gap cannot establish it without preparation-time assumptions. We consider this more seriously in the complementarities benchmark. Sufficiently large improvements in unobserved preparation can produce brokerage growth at constant costs outside this benchmark. The rejection therefore applies to the substitution model with the assumptions we state. Appendix C.4 gives the proof.

7.3. Pure priority: brokers obtain earlier approval

In the second benchmark, brokers provide no productive assistance. Both routes require identical frontline work, and brokerage only purchases priority at the approval stage. The approval facility completes the application already in service before selecting a waiting brokered application ahead of a direct one. Approval capacity, service requirements, and this priority rule remain unchanged after the staffing increase. The appendix formalizes the facility as a stationary queue with Poisson arrivals and identical exponential service requirements.

The common frontline process cancels from the recorded time gap. Under the specified priority

rule, the approval advantage satisfies

$$\Delta = g(Q, Q_B; \mu), \qquad g_Q > 0, \quad g_{Q_B} > 0. \tag{15}$$

Prediction 2. At a fixed approval facility, an increase in both total and brokered applications widens the recorded broker advantage. Pure priority therefore cannot explain expanding application flows and a narrowing recorded gap.

More applications raise congestion, and more priority applications increase the delay imposed on direct applicants. This result allows spare approval capacity and does not depend on how brokers or officials price priority. A faster common frontline process could still lower direct recorded time; it cannot reverse the gap prediction because both routes receive the same frontline improvement. The contradiction concerns priority as the sole broker service under the stated queueing assumptions. Appendix C.5 derives the result.

7.4. Productive complementarity

The third benchmark retains the assistance model's public technology and citizen choices. Its distinguishing feature is broker production. Brokers repeatedly interact with public workers to prepare, correct, and submit applications. When these interactions take less time, a broker can serve a client with less private effort. Let ω_p measure the time required for pre-submission interactions and e_m a broker's effort. A simple technology is

$$q_m = e_m \left(\frac{\xi}{\omega_p}\right)^{\psi}, \qquad c = c_0 + \bar{c}\left(\frac{\omega_p}{\xi}\right)^{\psi}, \qquad \xi, \bar{c}, \psi > 0, \quad c_0 \geq 0 \tag{16}$$

The cost expression follows when private effort costs $\bar{c}$ per unit and other costs are c_0 per client. Faster interactions lower broker cost; brokered clients' public work remains in the resource constraint. Holding c fixed recovers the substitution cost specification. Both benchmarks use the same preparation and recorded-time technologies.

Lower costs support a lower fee and encourage participation, counteracting the declining broker advantage. Direct applications can increase at the same time as relative time savings favor the direct route and easier access brings new citizens into the market.

Let $k = -d\log P / d\log F$ be the proportional fall in the broker fee. Direct applications and the

brokered share both increase when

$$\max\left\{0, \frac{V-P}{V}(y-x)\right\} < k < y, \qquad x \geq 0, \quad y > 0 \tag{17}$$

The lower bound makes the fee decline strong enough to raise broker share; the upper bound leaves direct application sufficiently more attractive to raise direct volume. Total applications then rise as well. Broker entry requires brokered volume to grow faster than applications per broker, which also adjust in equilibrium.

The time response imposes a separate requirement. In the assistance models, recorded time is $T_j = \ell_j \omega_s + d(Q;\mu)$, where $\ell_D > \ell_B > 0$ is post-submission frontline work, ω_s its elapsed time per unit, and $d(Q;\mu)$ common approval time. Faster frontline interactions narrow the gap. They also reduce direct recorded time if the frontline saving outweighs the approval congestion induced by additional applications. Brokered recorded time can change little when its smaller frontline saving approximately offsets the same increase in approval delay.

Prediction 3. If the broker-cost reduction supports (17), brokered volume grows faster than individual broker scale, and frontline time savings outweigh the increase in direct applicants' approval delay, productive complementarity generates the joint response: both application routes expand, broker share and entry rise, direct processing time falls, and the recorded gap narrows. Appendix C.6 gives the exact conditions and their derivation, including both public capacity constraints.

Public capacity can therefore reduce the assistance advantage of brokerage while increasing brokers' productivity. Under the stated conditions, that cost channel reconciles the joint responses that the competing benchmarks cannot. Brokers may also retain an approval-stage advantage; the priority result concerns an explanation based on priority alone.

8. Descriptive over of broker price quotes from mystery-shopper survey

The administrative records describe applications after formal submission, but do not show what citizens end up paying for brokers' private assistance or what that assistance includes. We complement these records with a descriptive mystery-shopper survey of brokers. The survey records initial prices, quoted completion times, assistance with missing documents, and descriptions of the public procedures involved. These observations help distinguish the services brokers offer from the outcomes recorded in the administrative data. They also show how large the quoted charges can be relative to a low daily wage. The survey describes offers and reported terms. We cannot observe completed transactions or identify how prices changed in response to the staffing reform.

8.1. Sample, scenarios, and measurement

We visit 160 brokers across 20 districts in Punjab, between September 18 and October 6, 2026. The interactions between surveyors and brokers aim to simulate the experience of an eligible person while obtaining a Scheduled Caste certificate. Eighty visits use a complete-document scenario and 80 use an incomplete-document scenario. The latter elicits whether a broker can help when a supporting family member's caste certificate is unavailable, i.e., the application package is incomplete.

The survey records an initial ordinary price and a quoted ordinary completion time for every visit. Follow-up questions cover whether the government fee is included, whether urgent service is available, help with the missing family certificate, reported terms in the event of rejection, and the possibility of applying at a Sewa Kendra without a broker. The instrument also records the broker's account of the Sewa Kendra's completion time and whether the broker offers service in another tehsil or district. Open-ended notes describe required documents, additional charges, and interactions with officials.

We distinguish an *initial ordinary quote* from a complete price for obtaining the certificate under the stated circumstances. Additional remarks sometimes identify separately charged paperwork, an unpriced prerequisite, or a condition that must be satisfied before the broker will proceed. Some incomplete-document interactions end with a refusal to proceed without the family certificate, despite a numerical price having been recorded earlier.

8.2. Quoted prices and the cost of access

Table 9 reports the distribution of initial ordinary quotes. The pooled median is INR 300 and the mean is INR 462.7. The interquartile range runs from INR 250 to INR 400, while the 90th and 95th percentiles are INR 700 and INR 1,500. Four visits record INR 3,500, the highest price in the sample.

The median initial quote is INR 300 in both document scenarios. The mean is INR 406.3 in complete-document visits and INR 519.1 in incomplete-document visits. The upper tail is also higher in the incomplete-document scenario. These differences combine variation across providers, places, and the terms attached to the quote. In particular, a low initial quote in the incomplete-document scenario may be conditional on obtaining an additional document elsewhere. It cannot automatically be interpreted as a lower complete cost of access.

What the price includes. In 154 of 160 visits (96.3 percent), the recorded response states that the government fee is included in the quote. Three responses state that it is separate and three are unclear. This makes the initial quote different from a net broker fee or markup. Government-fee inclusion also does not imply that all supporting-document costs are included: follow-up responses sometimes quote an additional charge for an affidavit or other paperwork.

Table 9. Initial ordinary prices and quoted completion times

	All visits	Complete	Incomplete
Visits	160	80	80
Panel A. Initial ordinary price (INR)			
Mean	462.7	406.3	519.1
Standard deviation	587.7	382.7	736.3
Minimum	100	100	100
25th percentile	250	250	218.8
Median	300	300	300
75th percentile	400	362.5	500
90th percentile	700	565	800
95th percentile	1,500	1,500	2,075
Maximum	3,500	2,250	3,500
Panel B. Quoted ordinary completion time (days)			
Mean	21.0	21.3	20.8
Standard deviation	15.8	18.0	13.3
Minimum	1	1	2
25th percentile	15	15	15
Median	20	20	20
75th percentile	25	25	25
90th percentile	30	30	30
95th percentile	40	35.3	40.3
Maximum	150	150	90

Notes: Mystery-shopper visits, September–October 2026. Complete and incomplete refer to the document scenario.

Prices relative to a daily wage. We compare broker quoted prices with Punjab's notified unskilled rural-employment wage of INR 360 per day, effective July 1, 2026.[8] The median quote represents 0.83 days of this wage. The 90th and 95th percentiles represent 1.94 and 4.17 days, respectively, while INR 3,500 corresponds to 9.72 days. These comparisons place the survey-period quotes and the wage benchmark in the same nominal units.

In 150 visits (93.8 percent), the initial quote exceeds half of the benchmark daily wage, INR 180. Forty-six quotes definitely exceed a full day's wage, and one range, INR 300–500, crosses that threshold. Thus the share exceeding one day's wage lies between 28.8 and 29.4 percent. Fifteen quotes (9.4 percent) exceed two days' wages. These amounts indicate a potentially substantial cash burden, even before any additional required expense.

8.3. What brokers offer and how they interact with the state

Assistance with missing documents. Out of 80 incomplete-document visits, brokers for 65% of surveyors offered help with missing family certificate (Details in Table 10).

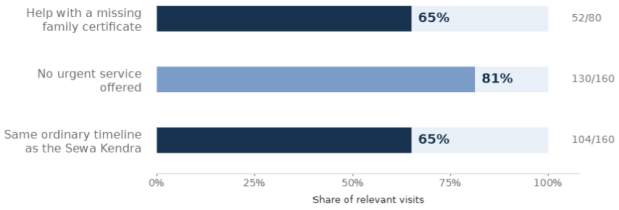
[8] Ministry of Rural Development, Lok Sabha Unstarred Question No. 235, answered July 21, 2026, Annexure II, Punjab row: INR 360 for 2026–27 under the Viksit Bharat–Guarantee for Rozgar and Ajeevika Mission (Gramin), or VB-G RAM G. The reply states that the revised rate took effect on July 1, 2026.

Figure 18. Broker quoted prices can raise distributional concerns



We ask surveyors to fill open-ended descriptions to understand nature of assistance more concretely. Brokers mention preparing an affidavit or stamp paper, supplying a form, and explaining which signatures or attestations the applicant needs.

Figure 19. Brokers reveal working with the State instead of circumventing it



Ordinary completion times and urgent offers. Brokers in most cases did not offer expedited services, and neither did they discredit the State. In 104 visits (65.0 percent), the broker's time quote or range is identical to the broker's report of the Sewa Kendra's completion time.

Taken together, the survey adds two types of evidence to the administrative analysis. First, documentation help and descriptions of public procedures show that brokers' offered services extend beyond a promise of faster approval. These observations support a framework where brokers and frontline public workers perform related tasks and interact during application preparation and submission. Second, the price distribution shows that initial private charges can be substantial relative to a low daily wage. Productive assistance and a potentially high cost to citizens can therefore coexist.

Table 10. Documentation assistance, urgent service, and responses to direct application

Response	Visits	Percent
Panel A. Missing-family-certificate help ($N = 80$)		
Help offered	52	65.0
No help offered	22	27.5
Unclear	6	7.5
Panel B. Terms among affirmative help responses ($N = 52$)		
Included in the quoted price	27	51.9
Additional charge	21	40.4
Conditional or no guarantee	4	7.7
Panel C. Urgent service ($N = 160$)		
Offered	29	18.1
Not offered	130	81.3
Unclear	1	0.6
Panel D. Response to applying directly at a Sewa Kendra ($N = 160$)		
Encouraged direct application	18	11.3
Neutral or factual response	88	55.0
Discouraged direct application	36	22.5
Not discussed	18	11.3

Notes: Panel A includes only incomplete-document scenarios. Panel B conditions on the 52 affirmative responses in Panel A. Panels C and D include all visits. Percentages use the panel-specific denominators. Panel D uses the categories recorded by the surveyor.

9. Discussion of citizen welfare

Citizen welfare depends on the full cost of obtaining a service. For a citizen with time cost θ, welfare in the model is

$$\mathcal{U}(\theta; F) = \max\{0,\ V - \theta(\tau + H),\ V - P - \theta\tau\}.$$

Under the complementarity conditions derived above, a marginal staffing increase lowers full direct application time, weakly lowers full brokered application time, and lowers the broker fee. Every citizen is therefore weakly better off. The previous route remains available at weakly lower cost, and citizens can change routes or begin applying. This welfare gain is compatible with an increase in broker use. The substitution benchmark can likewise improve welfare when full application times fall, despite predicting crowd-out.

Approval priority introduces distributional tradeoffs. Earlier approval for brokered applications can increase waiting time for direct applicants. At a fixed approval facility, priority creates no additional processing capacity, although allocating shorter waits to citizens with higher time costs can reduce the total cost of waiting. Broker fees also reduce citizen surplus, but the payments themselves are transfers to brokers. An assessment of net social welfare must additionally account

for the type of applicants who are unable to pay for broker services, real broker effort, operating costs, and the cost of public staffing.

The administrative evidence documents faster recorded processing for direct applications, which captures one component of the cost of access. It does not recover preparation-time or fee changes, so it is difficult to establish the model's full welfare conditions or quantify the change in citizen surplus. The welfare interpretation therefore remains conditional on these unobserved responses.

10. Conclusion

Better public services need not reduce citizens' use of private brokers. We study a staffing reform in a large Indian state that added workers to help citizens prepare and submit applications for government documents. These documents are often required to access public benefits and services. The reform expanded help at the application stage while leaving approval staffing essentially unchanged. Applications increase through both direct and brokered channels, but the increase is larger through brokers. The number of active brokers also increases. Direct applications are processed more quickly, narrowing the observed broker advantage. Brokerage therefore expands even as direct access improves.

In this paper, we see an instance of how public workers can make brokers cheaper to operate. Citizens need less private help when public staff can check documents, answer questions, and correct mistakes. Brokers also depend on these workers to prepare and submit applications for their clients. Easier access to public workers can therefore reduce the effort needed to serve each client. These lower costs translate to lower broker prices (as predicted in the model). This makes private assistance cheaper, encourage participation, and supports broker entry. Under the maintained restrictions, the substitution benchmark predicts a falling broker share. Priority alone predicts constant or higher processing-time gaps between direct and brokered applications as application flows rise at a fixed approval facility.

While we establish that complementarity can rationalize all of our empirics, we also show that better assistance at the application stage need not remove broker advantages at the approval stage. The broker advantage in approval by the statutory deadline falls substantially in lower-discretion services. It remains essentially unchanged in higher-discretion services. This pattern is consistent with brokers retaining an advantage where officials have more scope to influence outcomes. We do not have enough evidence from the administrative data to establish how that advantage is obtained[9].

[9]The mystery shoppers in our ongoing fieldwork are currently trying to ask this to brokers

Help with paperwork and repeated visits can reduce real costs. Access to priority can change who receives scarce official attention without increasing approval capacity. The expansion of brokerage therefore need not imply that every service brokers provide creates additional surplus.

The broader point of the paper is that the cost of accessing the state depends on both public services and the private markets that form around them. Broker prevalence alone does not show how well/poorly the state serves citizens. The welfare consequences depend on what brokers provide, what citizens pay in money and time, and the distributional consequences of such an unregulated market. Administrative burdens therefore can create private markets for overcoming them, and those markets respond to changes in the state. Building state capacity therefore requires understanding both how government services are produced and how private markets adjust around them.

A. Results with a post-reform indicator

Table 11 reports OLS and IV estimates that include a post-reform indicator alongside district fixed effects. The indicator equals one from March 10, 2023 onward. It allows a common shift in outcomes at the reform date that is separate from the estimated staffing response.

Application volume and active brokerage continue to increase with staffing. The IV coefficients are 0.220 for total applications per capita, 0.050 for brokered applications per capita, and 0.170 for direct applications per capita. Active brokers per capita also increase, with an IV coefficient of 0.034. All four estimates are positive and statistically significant, as are their OLS counterparts. The expansion in broker-mediated applications therefore continues to coexist with an expansion in direct applications.

Processing times retain the main pattern. For an increase of ten frontline workers per 100,000 residents, the IV estimates imply a decline of 0.524 days in mean processing time. Direct processing time falls by 0.63 days, while brokered processing time falls by an imprecisely estimated 0.14 days. The direct-minus-broker gap narrows by 0.49 days. Thus, the narrowing gap continues to reflect faster direct processing rather than slower brokered processing. The estimated effect on the delay rate is small and statistically indistinguishable from zero.

The broker-share result is less robust. Its OLS coefficient remains positive at 0.032 (standard error 0.013), but its IV coefficient is close to zero and slightly negative at -0.005 (standard error 0.039). The IV estimate therefore differs from the positive baseline result in both its point estimate and its precision. This specification supports an increase in broker activity alongside improved direct processing, but does not establish that brokered applications expand disproportionately. The reported first-stage statistics are approximately 11.4 across outcomes.

Table 11. Staffing results with a post-reform indicator

Outcome	OLS	IV	Pre-reform mean	N	KP F (IV)
A. Applications and brokerage					
Total applications per capita	0.141***	0.220***	10.975	6,408	11.404
	(0.024)	(0.044)			
Brokered applications per capita	0.036***	0.050**	2.282	6,408	11.404
	(0.008)	(0.014)			
Direct applications per capita	0.105***	0.170***	8.693	6,408	11.404
	(0.018)	(0.032)			
Broker share (percentage points)	0.032*	−0.005	19.654	6,408	11.404
	(0.013)	(0.039)			
Active brokers per capita	0.024***	0.034***	1.082	6,408	11.404
	(0.005)	(0.007)			
B. Processing times and delays					
Mean processing time (days)	−0.0348**	−0.0524*	14.2939	6,408	11.4041
	(0.0110)	(0.0197)			
Delay rate (share)	−0.0004	−0.0003	0.3445	6,408	11.4041
	(0.0005)	(0.0007)			
Direct-minus-broker processing time (days)	−0.034***	−0.049**	3.741	6,266	11.353
	(0.006)	(0.015)			
Brokered processing time (days)	−0.007	−0.014	11.122	6,266	11.353
	(0.010)	(0.010)			
Direct processing time (days)	−0.041**	−0.063**	14.863	6,266	11.353
	(0.012)	(0.021)			
District fixed effects	Yes	Yes			
Post-reform indicator	Yes	Yes			

Notes: Each row reports the coefficient on frontline staffing from separate OLS and IV regressions. Staffing is measured as frontline workers per 100,000 residents. Standard errors are in parentheses. All specifications include district fixed effects and an indicator equal to one from March 10, 2023 onward. Application and active-broker outcomes are reported per capita; broker share is measured in percentage points and the delay rate as a fraction. Processing times and the direct-minus-broker gap are measured in days. N is the number of district-date observations; the sample size and reported pre-reform mean are the same for OLS and IV within each row. KP F reports the Kleibergen–Paap first-stage statistic for the IV regression. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

B. Robustness

Figure 20. District exposure and pre-shock service delivery outcomes

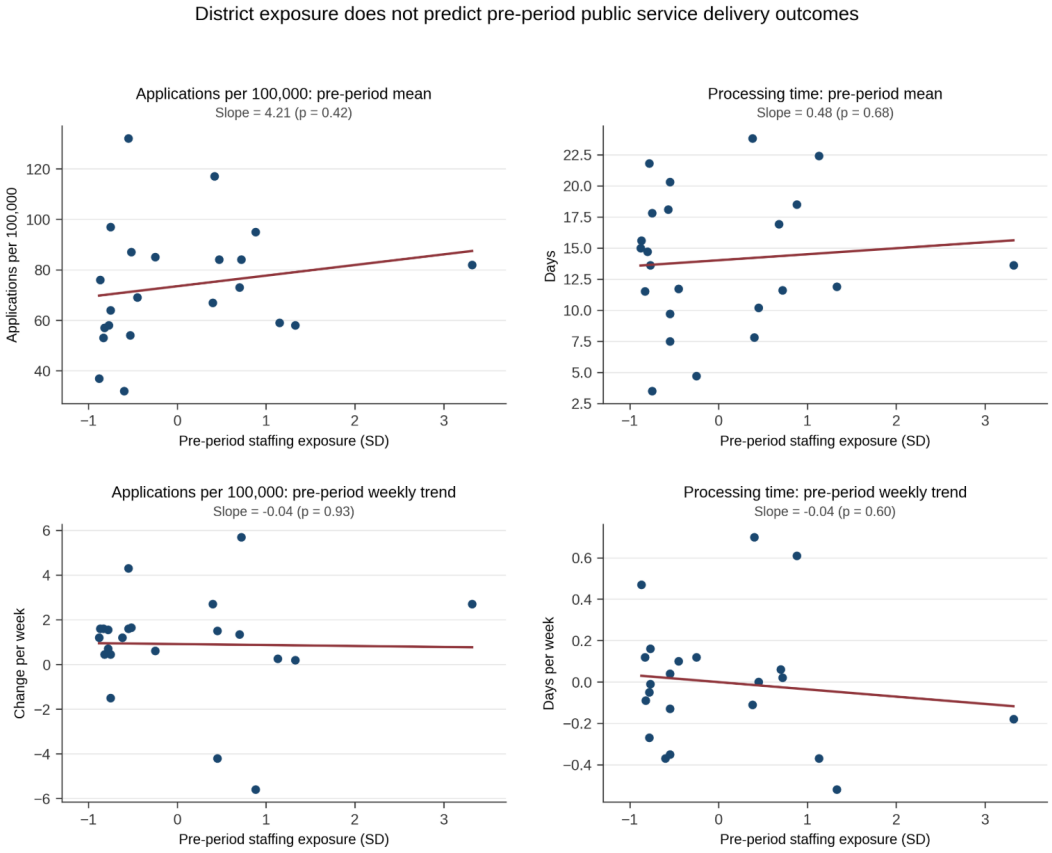
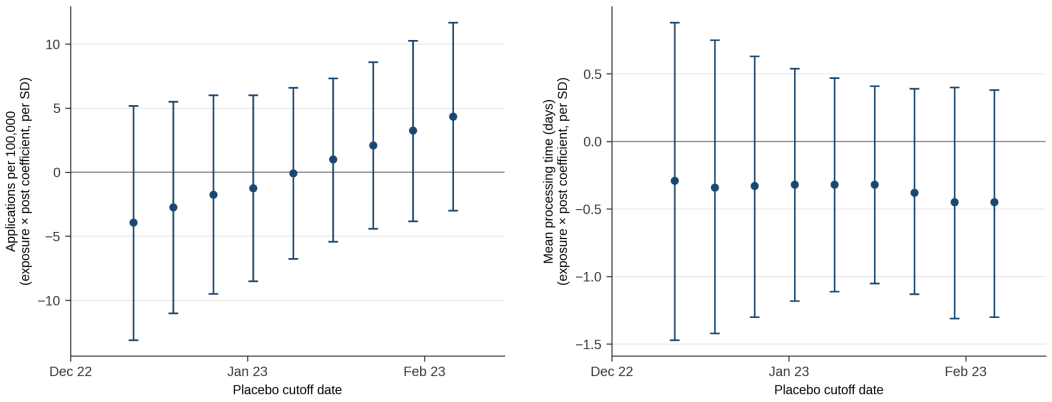


Figure 21. Placebo reform dates in pre-reform period for service delivery outcomes



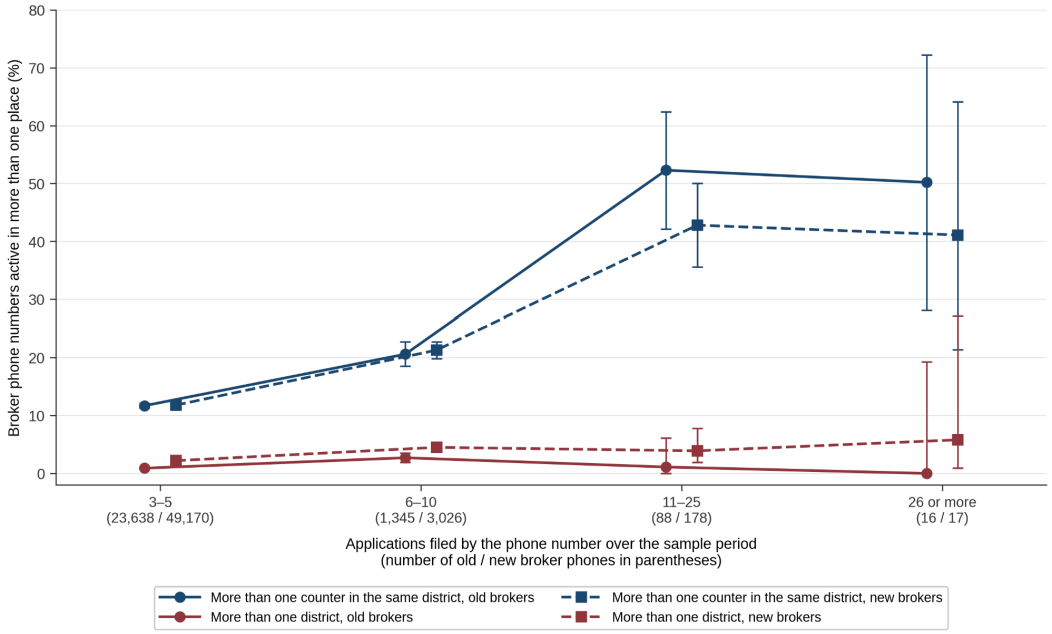
Notes: Each point is the coefficient on exposure × post from a separate regression that uses pre-reform data only. "Post" equals one for dates on or after the placebo cutoff shown on the horizontal axis. Placebo cutoffs are set weekly from Dec 2022 to Feb 2023. Exposure is district pre-reform frontline staff per 100,000 residents, standardized to mean zero and unit standard deviation. All regressions include district and day-of-week fixed effects. Bars show 95% confidence intervals based on standard errors clustered by district and date. The left panel reports applications per 100,000 residents and the right panel reports mean processing time in days.

Table 12. Broker-threshold sensitivity check

Minimum applications per phone	3	4	6	8	10
All applications per 100,000	0.188	0.188	0.188	0.188	0.188
	(0.025)	(0.025)	(0.025)	(0.025)	(0.025)
District-date observations	6,569	6,569	6,569	6,569	6,569
Mean processing time (days)	-0.043	-0.043	-0.043	-0.043	-0.043
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
District-date observations	6,569	6,569	6,569	6,569	6,569
Direct minus broker time (days)	-0.041	-0.033	-0.025	-0.006	-0.002
	(0.009)	(0.007)	(0.005)	(0.013)	(0.021)
District-date observations	6,438	6,149	4,694	2,980	2,063
Broker processing time (days)	-0.010	-0.010	-0.016	-0.040	-0.052
	(0.008)	(0.008)	(0.012)	(0.024)	(0.031)
District-date observations	6,439	6,149	4,694	2,980	2,063
Direct processing time (days)	-0.051	-0.046	-0.044	-0.043	-0.043
	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)
District-date observations	6,568	6,569	6,569	6,569	6,569
Broker share (percentage points)	0.065	0.059	0.025	0.014	0.008
	(0.019)	(0.010)	(0.006)	(0.004)	(0.003)
District-date observations	6,408	6,408	6,408	6,408	6,408
Active broker phones per 100 applications	0.092	0.066	0.027	0.014	0.009
	(0.010)	(0.006)	(0.004)	(0.003)	(0.002)
District-date observations	6,546	6,546	6,546	6,546	6,546
District and day-of-week FE	Yes	Yes	Yes	Yes	Yes

Notes: Each cell is a separate 2SLS regression, with standard errors clustered by district and application date in parentheses. Coefficients are per additional staff member per 100,000 residents. We reproduce outcomes of Tables 5–7.

Figure 22. Broker mobile numbers active in more than one location, by application volume



Notes: Old brokers are numbers whose first application is before 10 March 2023. Numbers are grouped by the total number of applications they filed over the sample period. The blue series shows the percent of phones that filed applications at more than one SK counter within a single district. The red series shows the percent that filed applications in more than one district. Bars show 95% confidence intervals.

C. Deriving a model of citizen choice, broker competition, and state capacity

This appendix derives citizen demand, broker competition and entry, and the three benchmark predictions. The results concern a differentiable branch of feasible stationary equilibria. A dot denotes a total derivative with respect to $z = \log F$, holding approval capacity and all other primitives fixed. Thus $\dot{Y} = dY/d\log F$ and $\varepsilon_Y = \dot{Y}/Y = d\log Y/d\log F$.

C.1. Citizen choice and broker competition

Citizen cutoffs. From (12), direct application and brokerage provide equal utility at $\theta_1 = P/H$. Brokerage and nonparticipation provide equal utility at $\theta_2 = (V - P)/\tau$. Assume

$$0 < \theta_1 < \theta_2 < \bar{\theta}. \tag{18}$$

Below θ_1, direct application dominates brokerage and gives positive utility. Between θ_1 and θ_2, brokerage dominates direct application and gives positive utility. Above θ_2, neither route gives positive utility. Integrating the constant density λ over these intervals gives (13). In particular,

$$1 - S_B = \frac{Q_D}{Q} = \frac{P\tau}{H(V - P)}. \tag{19}$$

Inverse demand. Holding the public delay forecasts (τ, H) fixed, define

$$h = \frac{1}{\tau} + \frac{1}{H}, \qquad a = \frac{VH}{\tau + H}. \tag{20}$$

Citizen demand implies $Q_B = \lambda h(a - P)$ and hence $P = a - Q_B/(\lambda h)$. Broker m chooses q_m, taking its competitors' quantities and district-wide public delays as given, to maximize

$$\pi_m = (P - c)q_m - K. \tag{21}$$

The broker internalizes its effect on P. It does not internalize the effect of its individual quantity on district-wide congestion. This is a maintained approximation for a market with many brokers; aggregate delay forecasts must nevertheless be correct in equilibrium.

Symmetry and free entry. The first-order condition and the zero-net-profit condition are

$$P - c = \frac{q}{\lambda h}, \qquad (P - c)q = K. \tag{22}$$

Writing $r = P - c$ for the markup, these imply

$$r = \sqrt{\frac{K}{\lambda h}}, \qquad P = c + r, \qquad q = \sqrt{\lambda h K}, \qquad M = \frac{a - c}{r} - 1, \qquad Q_B = Mq. \tag{23}$$

Conditional profit is strictly concave in own quantity because inverse demand has slope $-1/(\lambda h)$. Entry is treated as continuous, so M measures active broker units. An interior broker market requires $a > c + r$; at least one full broker requires $a - c \geq 2r$. The citizen-support restriction in (18) must also hold. At $a \leq c + r$, the interior formulas do not describe an active broker market. With an integer number of brokers, entry may occur in steps rather than through a smooth local response.

C.2. Application processing time and equilibrium

The common assistance environment. Let $b_j \geq 0$ and $\ell_j > 0$ denote frontline work before and after formal submission. Work per application is $\phi_j = b_j + \ell_j$, and

$$L = \phi_D Q_D + \phi_B Q_B < F, \qquad Q_D + Q_B < \mu. \tag{24}$$

Let $\omega_p(F, L)$ and $\omega_s(F, L)$ be elapsed time per unit of frontline work before and after submission. Both are differentiable, increasing in workload, and decreasing in staffing when workload is held fixed. These are reduced-form interaction and delay technologies. They count elapsed time rather than additional staff-time allocations. The two clocks are

$$R_j = \bar{R}_j + b_j \omega_p(F, L), \qquad T_j = \ell_j \omega_s(F, L) + d(Q; \mu), \qquad d_Q > 0, \tag{25}$$

where $\bar{R}_j \geq 0$ is fixed preparation time and d includes approval waiting and service. The assistance benchmarks maintain $\ell_D > \ell_B$, $b_D \geq b_B$, and $\bar{R}_D \geq \bar{R}_B$. Consequently,

$$\begin{aligned} \tau &= \bar{R}_B + b_B \omega_p + \ell_B \omega_s + d(Q; \mu), \\ H &= (\bar{R}_D - \bar{R}_B) + (b_D - b_B)\omega_p + (\ell_D - \ell_B)\omega_s, \\ \Delta &= (\ell_D - \ell_B)\omega_s. \end{aligned} \tag{26}$$

Substitution and complementarity use the same specifications for both clocks and public work. Only the broker-cost technology differs. No preparation-time improvement is allowed exclusively in the complementarity benchmark.

Equilibrium consistency. For candidate flows (Q_D, Q_B), compute L, the public delays, (τ, H), c, and the fee in (23). Equilibrium requires the candidate flows to equal citizen demand in (13). Define these two excess-flow equations by

$$E_D = Q_D - \lambda P / H = 0, \qquad E_B = Q_B - \lambda\Big[(V - P)/\tau - P/H\Big] = 0. \tag{27}$$

Assume the Jacobian $J = \partial(E_D, E_B)/\partial(Q_D, Q_B)$ is nonsingular at the equilibrium under consideration. The implicit-function theorem then gives the local response

$$\begin{pmatrix} \dot{Q}_D \\ \dot{Q}_B \end{pmatrix} = -J^{-1} \frac{\partial(E_D, E_B)}{\partial z}. \tag{28}$$

All time and fee responses used below are evaluated along this equilibrium branch. For example,

$$\dot{\omega}_v = F \frac{\partial \omega_v}{\partial F} + \frac{\partial \omega_v}{\partial L} \dot{L}, \qquad \dot{L} = \phi_D \dot{Q}_D + \phi_B \dot{Q}_B, \quad v \in \{p, s\}. \tag{29}$$

More staffing need not lower equilibrium delays when induced workload is sufficiently large. This is why the positive result imposes conditions on the total delay responses, rather than just their partial derivatives. Strict inequalities in (24) and the interior conditions remain valid for sufficiently small staffing changes on a continuous equilibrium branch. The local results do not apply to an overloaded approval facility with $Q \geq \mu$.

C.3. Useful equilibrium derivatives

Define

$$x = -\frac{\dot{\tau}}{\tau}, \qquad y = -\frac{\dot{H}}{H}, \qquad k = -\frac{\dot{P}}{P}, \qquad \nu = \frac{\dot{h}}{h} = \frac{H}{\tau + H} x + \frac{\tau}{\tau + H} y. \tag{30}$$

Because K and λ are fixed, the broker-market formulas give

$$\frac{\dot{r}}{r} = -\frac{\nu}{2}, \qquad \frac{\dot{q}}{q} = \frac{\nu}{2}, \qquad k = -\frac{\dot{c}}{P} + \frac{r}{2P} \nu. \tag{31}$$

Differentiating citizen demand and (19) yields

$$\begin{aligned} \varepsilon_{Q_D} &= y - k, \\ \varepsilon_Q &= x + \frac{P}{V - P} k, \\ \dot{S}_B &= (1 - S_B) \left[x + \frac{V}{V - P} k - y \right]. \end{aligned} \tag{32}$$

These identities include participation and the changing broker fee. Route volumes, total applications, and broker share are linked by accounting and are not independent restrictions.

C.4. Proof of the substitution prediction

With $\dot{c} = 0$, equation (31) gives $k = r\nu/(2P)$. Define

$$w = \frac{H}{\tau + H}, \qquad \zeta = \frac{Vr}{2P(V - P)}. \tag{33}$$

The sign of $\dot{S}_B$ is the sign of

$$(1+\zeta w)x - \Big[1-\zeta(1-w)\Big]y. \tag{34}$$

Since $c \geq 0$, $r \leq P$. An active broker market also implies $P < a = Vw$, so

$$\zeta < \frac{1}{2(1-w)}. \tag{35}$$

In particular, $1-\zeta(1-w) > 0$. For $y > 0$, the exact threshold for crowd-out is therefore

$$\frac{x}{y} < \frac{1-\zeta(1-w)}{1+\zeta w}. \tag{36}$$

The fraction on the right decreases strictly with ζ. Using its upper bound gives

$$\frac{1-\zeta(1-w)}{1+\zeta w} > \frac{1-w}{2-w} = \frac{\tau}{2\tau+H}. \tag{37}$$

Hence the weak bound in (14) implies strict crowd-out: $\dot{S}_B < 0$. This proves Prediction 1. The result allows the markup and participation to adjust. It is a sufficient condition for a declining share, not a claim that brokered volume must fall under every improvement in public assistance.

The unobserved clock qualification follows from

$$H = (R_D - R_B) + \Delta, \qquad \tau = R_B + T_B. \tag{38}$$

Recorded processing-time responses do not determine x/y without assumptions on R_D and R_B. If common access improves sufficiently relative to H, the expression in (34) can become positive with constant costs. The analytical rejection is therefore conditional on the full-time restriction, not established by the recorded gap alone.

C.5. Proof of the pure priority prediction

Let $B = Q_B$ and $D = Q_D$, so $Q = B + D$. Both routes require identical public work. At the approval facility, arrivals from each route are independent Poisson processes, service times are exponential with common mean $1/\mu$, and $Q < \mu$. Service is nonpreemptive: an application already in service is completed before the next one is chosen. Waiting brokered applications have strict priority, with first-in-first-out service within each route.

Deriving the waiting times. An arrival sees an expected residual service time, including the possibility that the facility is idle, of $W_0 = Q/\mu^2$. This equals utilization Q/μ times the exponential distribution's mean residual service time $1/\mu$. Let W_B and W_D denote mean waiting times before service. A brokered arrival waits for residual service and brokered applications already waiting.

Little's law gives

$$W_B = W_0 + \frac{B}{\mu} W_B, \qquad W_B = \frac{Q}{\mu(\mu - B)}. \tag{39}$$

A direct arrival also waits for direct applications already ahead of it and brokered applications that arrive during its wait. Thus

$$W_D = W_0 + \frac{B}{\mu} W_B + \frac{D}{\mu} W_D + \frac{B}{\mu} W_D, \qquad W_D = \frac{Q}{(\mu - B)(\mu - Q)}. \tag{40}$$

Adding the common service time $1/\mu$ and subtracting gives

$$g(Q, B; \mu) = W_D - W_B = \frac{Q^2}{\mu(\mu - B)(\mu - Q)}. \tag{41}$$

This is the two-class exponential-service specialization of the nonpreemptive priority formula in Larson and Odoni, *Urban Operations Research* (1981), Section 4.9.1.

Comparative statics. For $0 < B < Q < \mu$,

$$g_Q = g\left(\frac{2}{Q} + \frac{1}{\mu - Q}\right) > 0, \qquad g_B = \frac{g}{\mu - B} > 0. \tag{42}$$

Since the common recorded frontline time cancels from the gap,

$$\dot{\Delta} = g_Q \dot{Q} + g_B \dot{B} > 0 \quad \text{if } \dot{Q} > 0 \text{ and } \dot{B} > 0. \tag{43}$$

This proves Prediction 2 independently of the broker fee or any payment for priority. Citizen choice and broker entry may determine which flow changes occur; they cannot change this implication conditional on those flows.

Priority reallocates the same approval resource. In fact, the average approval time across both routes satisfies

$$\frac{B(1/\mu + W_B) + D(1/\mu + W_D)}{Q} = \frac{1}{\mu - Q}. \tag{44}$$

No additional approval capacity has been assigned to brokers. The result allows utilization strictly below one. Fixed approver headcounts alone are not sufficient: effective service capacity, service requirements, arrival assumptions, and the priority rule are also maintained. A changing priority rule or a productive broker service would constitute a different benchmark.

C.6. Derivation of the complementarity result

The productivity channel. From (16), delivering q_m applications requires $e_m = q_m(\omega_p/\xi)^\psi$ units of broker effort. The cost of this effort plus other per-client costs is $[c_0 + \bar{c}(\omega_p/\xi)^\psi]q_m$, giving

the stated marginal cost. Its equilibrium response is

$$\dot{c} = \psi(c - c_0)\frac{\dot{\omega}_p}{\omega_p}. \tag{45}$$

Thus a fall in total pre-submission interaction time lowers broker cost. The public work $b_B Q_B$ remains in (24); the cost saving concerns private effort, not an uncounted supply of public labor.

To separate the three forces, substitute $P = c + r$ into broker demand and differentiate:

$$\dot{Q}_B = -\lambda\frac{V - c - r/2}{\tau^2}\dot{\tau} + \lambda\frac{c + r/2}{H^2}\dot{H} - \lambda h\dot{c}. \tag{46}$$

The first term is the participation response to common access, the second is the response to the assistance advantage, and the third is the broker supply response. The cost term is absent under substitution. All three terms use total equilibrium derivatives.

Applications and broker share. Suppose $x \geq 0$ and $y > 0$. From (32), direct volume rises if $k < y$, and broker share rises if $k > (V - P)(y - x)/V$. Requiring $k > 0$ also ensures that total applications rise. These are precisely the conditions in (17). Since $Q_B = S_B Q$, brokered volume then rises too.

The interval is nonempty for every $x \geq 0$ and $y > 0$: its lower bound is strictly below y because $(V - P)/V < 1$. Under the substitution bound, the constant-cost fee response is too small to cross the share-growth threshold. Equation (31) shows how a fall in c can raise k above that threshold. A sufficiently large fee decline can also violate $k < y$, however; lower broker costs alone do not guarantee increased direct volume.

Entry. Let $s = S_B$ and $\beta = P/(V - P)$. Accounting and (32) give

$$\varepsilon_{Q_B} = \frac{x - (1 - s)y + (\beta + 1 - s)k}{s}, \qquad \varepsilon_M = \varepsilon_{Q_B} - \frac{\nu}{2}. \tag{47}$$

Hence entry rises exactly when

$$\varepsilon_{Q_B} > \frac{1}{2}\left[\frac{H}{\tau + H}x + \frac{\tau}{\tau + H}y\right]. \tag{48}$$

Equivalently, $k > k_E$, where

$$k_E = \frac{s\nu/2 - x + (1 - s)y}{\beta + 1 - s}. \tag{49}$$

The application and entry conditions can therefore be written together as

$$\max\left\{0, \frac{V - P}{V}(y - x), k_E\right\} < k < y. \tag{50}$$

This interval is nonempty if $k_E < y$, or equivalently $x + \beta y > s\nu/2$. That additional restriction makes explicit why entry is not automatic. It can hold within the substitution restriction: holding τ, H and x/y fixed, a sufficiently small positive baseline broker share makes the right-hand side small, while $\beta = (1-s)H/\tau$ remains positive. This demonstrates a nonempty region of market-response conditions; the public delay technology must also satisfy the timing conditions below.

Recorded time and both public constraints. Because frontline work per application is fixed,

$$\dot{\Delta} = (\ell_D - \ell_B)\dot{\omega}_s, \qquad \dot{T}_D = \ell_D \dot{\omega}_s + d_Q \dot{Q}, \qquad \dot{T}_B = \ell_B \dot{\omega}_s + d_Q \dot{Q}. \tag{51}$$

The gap narrows if $\dot{\omega}_s < 0$. Direct applications become faster if

$$d_Q \dot{Q} < -\ell_D \dot{\omega}_s. \tag{52}$$

These conditions explicitly retain the extra approval congestion from $\dot{Q} > 0$. Brokered time is unchanged to first order when $d_Q \dot{Q} = -\ell_B \dot{\omega}_s$. At that equality, direct time still falls because $\ell_D > \ell_B$.

Prediction 3 follows by combining (50), $\dot{\omega}_p < 0$, $\dot{\omega}_s < 0$, and (52) on the interior, feasible equilibrium branch defined by (27). These are sufficient conditions for the joint response. They are not independent assignments of fees, flows, and times: all derivatives must come from that same equilibrium, with workload and approval congestion included as in (29). The analytical comparison identifies the cost force that can overturn crowd-out; it does not infer the unobserved preparation process from recorded times.

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